You have been taught science for several years now. You have learned the names of molecules and the stages of cell division and the laws of inheritance and the structure of DNA. You have written these things in examinations and, in many cases, you have written them correctly. Your teachers have told you that you are learning science.

But there is something that nobody has told you, and it is this: the facts you have memorised are not science, they are the residue of science, what remains after a human being has spent months or years in genuine uncertainty, asking a question that nobody had answered before, designing ways to test possible answers, finding that most of the possible answers were wrong, and eventually arriving at something true. The facts in your textbook are the final sentences of stories that your textbook never told you.

This book tells you six of those stories, chosen because together they form a single argument that runs across one hundred and thirteen years of biology. The argument is not complicated, but it is not obvious either, and by the time you finish this book you will have arrived at it yourself rather than having it handed to you, which is the only way it will mean anything.

The six stories involve a mathematician who solved a biology problem in an afternoon and was slightly embarrassed by how simple the solution was; two scientists who proposed the structure of DNA in nine hundred words and buried their most important claim in a single quiet sentence near the end of the paper; two young researchers in California who designed what many scientists have called the most beautiful experiment in biology; a physicist in Chennai who mapped the geometric boundaries of what proteins are permitted to be; a team of engineers in London who taught a computer to read biological information in a way that no human mind can fully explain or reproduce; and a historian who went back to ask why something so mathematically obvious had taken the scientific community so long to see.

What connects these six stories is not the subject matter, though the subject matter is related across all of them. What connects them is the question that each story is really answering, underneath the specific biology: how do you find something true in a world that gives you very little to go on, and what kind of thinking does that require? These are not questions your examination will ask you. They are more important than any question your examination will ask you, because they determine whether you become someone who uses knowledge or merely someone who carries it from a textbook to an answer sheet without anything happening in between.

You do not need to be at a famous institution to read this book. You do not need a laboratory or a supervisor or access to an academic journal. You need the six papers, which are reproduced here in full, and you need the willingness to sit with a difficult idea long enough for it to become clear, rather than reaching immediately for a definition you can copy into a notebook and memorise before the examination.

The papers were written by human beings who did not know the answers when they started. That is the most important thing to understand about them. They were written in the middle of confusion, not at the end of it. Reading them the way this book will show you means recovering that confusion, feeling the full weight of the question before you feel the relief of the answer. It means reading slowly, which is the opposite of how examinations have trained you to read. It means asking, at every step, not only what this sentence says but how the person who wrote it could possibly have known that it was true.

If you read this book in that spirit, something will happen to you that no examination can measure and no certificate can record. You will begin to understand what science actually is, as opposed to what it looks like from the outside when all the confusion has been cleared away and only the clean facts remain. And you will find, very likely to your own surprise, that you are capable of exactly the kind of thinking that produced those facts in the first place, because that capacity was always in you and this book is simply the first thing that has asked for it.

Preface

Why I Wrote This From Kuchinda

This book began with a problem I could not solve.

I was a BSc student, working through Dunn and Sinnott's genetics textbook on my own, not because anyone asked me to but because the questions in it interested me more than the answers I was being given in class. Before the chapter on population genetics, there was a problem about an island. A population of plants with flowers of two colours, one colour dominant over the other, present in certain proportions. The question was what would happen to those proportions after a few generations of random mating. I spent hours on it. I could not find the answer. Not because I was not trying, but because I did not yet have the framework that makes the answer visible.

What I did not know at the time was that Dunn and Sinnott had placed that problem there deliberately. They put it before the Hardy-Weinberg chapter because they wanted to test whether the reader could arrive at the logic that Hardy and Weinberg had arrived at, before being shown the framework. They were practising a pedagogy of withheld answers, trusting that the experience of genuinely not being able to solve a problem would make the solution, when it arrived, land as a discovery rather than a delivery.

When I eventually encountered Hardy-Weinberg equilibrium, the problem dissolved in minutes. The answer was simple. It had always been simple. But it was completely invisible without the framework, and the framework had never been given to me as a framework. It had been given to me as an equation to memorise and apply. Nobody had shown me where the equation came from, what confusion it dissolved, what kind of mind produced it and how. Nobody had shown me the structure beneath.

That experience stayed with me for twenty years. It is the reason this book exists.

I am not a historian of science. I am not a philosopher of science. I am a zoologist at Kuchinda Degree College in Odisha, affiliated to Sambalpur University, where I teach genetics and ecology, supervise research students, and do fieldwork on earthworm populations in mining-affected landscapes. I wrote this book not to make an argument but to understand something, something I had been circling for three decades without quite being able to name.

The something is this: scientific thinking is not the same as scientific knowledge, and the difference between them is almost never shown to students. What students are shown is the knowledge, the equations, the results, the names attached to principles and laws and plots. What they are not shown is the thinking that produced the knowledge, the confusion before the clarity, the question before the answer, the human being working through a problem with whatever tools their training had given them, in whatever place and time they happened to inhabit, toward a result they could not yet see.

I wanted to see that thinking. Not described abstractly, not summarised in a history of science textbook, but visible in the actual words of the actual papers, the way a carpenter's thinking is visible in the joints of a piece of furniture if you know where to look. I wanted to sit with Hardy's two pages and understand not just what he concluded but how he saw what Yule and Pearson could not see. I wanted to read Watson and Crick's nine hundred words slowly enough to find the sentence that contains more than it announces. I wanted to understand what Ramachandran was doing geometrically in Chennai in 1963 and why his work became invisible infrastructure for a neural network built in London fifty-eight years later. I wanted to read Darwin's argument for natural selection and understand why it is different in kind from every other paper in this collection, why it is the one paper that needed no mathematics, no physics, no computation, only a naturalist's patience and a mind willing to follow biological observation wherever it led.

This book is the record of that understanding, or the beginning of it. It is organised around eight texts spanning one hundred and fifty-eight years of biology, from Darwin's On the Origin of Species in 1859 to the AlphaFold paper in 2021. Each chapter reads one text carefully, not for the facts it contains but for the thinking it performs. The chapters build on each other in a way that I did not plan in advance but that became visible as I wrote, a progression from mathematical reading to structural reading to experimental reading to geometric reading to computational reading to biological reading, each chapter demonstrating a different way that a human mind can approach the question of what the biological world actually is and how it can be known.

Along the way, the book arrives at an argument I did not set out to make. The argument is that biological information is readable at every level of organisation, from the population to the molecule to the protein backbone to the sequence, and that this readability is not a coincidence but a product of natural selection, which has been structuring biological systems into information-rich readable forms for billions of years. I did not write this book to make that argument. I wrote it to understand something for myself, and that argument is what the understanding produced. Whether it is correct is for others to judge. But it felt, when I arrived at it, like the thing I had been trying to see since I spent those hours on the island problem and could not find the answer.

I want to say one thing about where this book was written.

Kuchinda is a small town in Sambalpur district in Odisha. Kuchinda Degree College is not an institution whose name is known outside its district. I have no laboratory with advanced equipment, no large research group, no affiliation with any of the institutions that the history of science has tended to treat as the places where important thinking happens. I mention this not as a disclaimer but because I think it is relevant to what the book is trying to do.

The papers in this book were written at Cambridge, at Caltech, at the University of Madras, at DeepMind in London, at the Cavendish Laboratory. With one exception, Ramachandran in Chennai, they were written at institutions that the scientific world immediately recognises as places where important work is done. But the thinking in those papers is not the property of those institutions. It belongs to anyone who reads the papers carefully enough to see it. And the need to read them carefully, to recover the thinking that the textbook presentation conceals, is felt most acutely not at the famous institutions, where students are surrounded by people who do science every day, but at places like Kuchinda, where a student who wants to understand the structure beneath the facts has to find it largely alone.

This book is for that student. It is also for the teacher who has been trying, for years, to show students the structure beneath, and who has not been able to find the words because nobody has written them down in a form that can be handed to a class 12 student or a BSc student or a curious teacher on a Sunday afternoon in a small town anywhere in India or anywhere in the world where science is taught as a body of knowledge rather than as a way of thinking.

I have tried to write those words. Whether I have succeeded is not for me to say. But the attempt was made from Kuchinda, which is exactly where it should have been made, because Kuchinda is where the need for it is most clearly visible.

The papers are waiting. Let us read them.

Alok Patel
Kuchinda Degree College, Sambalpur University Odisha, India

A Note on Reading These Texts

The papers in this book were written for specialists. Hardy's letter was addressed to biologists who already knew what a dominant allele was. Watson and Crick's paper was addressed to chemists and biophysicists who already knew what an X-ray diffraction pattern looked like. Ramachandran's paper was addressed to structural biologists who already understood what a peptide bond was and why its geometry mattered. This does not mean that you, reading as a non-specialist, are in the wrong place. It means that reading these papers as a non-specialist requires a particular kind of preparation, and this note is that preparation.

The first thing to understand is that you are not reading these papers for information. You have already been given the information, in your biology and chemistry classes. You know what Hardy-Weinberg equilibrium is, in the sense that you can state the equation and apply it to a numerical problem. You know what the double helix is, in the sense that you can draw it and name its components. You know, in outline, what DNA replication and protein structure involve. The papers will not give you information you do not have. They will show you the thinking that produced the information you already have, which is a completely different thing and a much more valuable one.

The second thing to understand is that reading a scientific paper slowly is not the same as reading it carefully. Reading slowly without a question in mind produces nothing except a slow accumulation of sentences. What this book asks you to do is to hold the question the paper is trying to answer clearly in your mind before you begin reading, and to read each sentence by asking two things: what is being claimed here, and how could the person who wrote this sentence have known that it was true? These two questions, applied consistently, will reveal the structure of the argument in a way that no amount of slow reading without them will produce. The chapters in this book are designed to give you the question before you encounter the paper, so that when you arrive at the paper you already know what to look for.

The third thing to understand is that the technical vocabulary in these papers, the notation in Hardy's algebra, the crystallographic terminology in Watson and Crick, the geometric notation in Ramachandran, is not an obstacle to reading them but a signal about the kind of thinking being done. When Hardy writes a probability expression, he is doing something precise and rigorous that ordinary language cannot do as cleanly. When Watson and Crick describe the antiparallel orientation of the DNA strands, they are compressing into a single technical phrase a geometric relationship that would take several sentences of ordinary prose to describe. The vocabulary is dense because the thinking is precise. Your task as a reader is not to master the vocabulary before you read, but to use the chapters in this book to understand what the vocabulary is doing, which is carrying precision that the argument requires. Where the vocabulary becomes a genuine barrier, the chapters will walk you through it. Where it does not, allow it to be there without letting it stop you from following the argument that surrounds it.

These papers were written by human beings who were confused, who made mistakes, who worked with incomplete information, and who were operating under institutional and social conditions that shaped what they could see and what they could say. The chapters will draw your attention to those conditions where they matter. Read the papers with that in mind. Read them as records of human minds at work in specific places at specific times, not as tablets handed down from an authoritative height. The authority in these papers is earned, not granted, and understanding how it is earned is the purpose of this book.

Chapter 1

The Afternoon Hardy Solved a Problem He Found Embarrassing

G. H. Hardy, 1908

Most of us who have studied genetics were given the Hardy-Weinberg equation before we were given the question it answers. We were shown $p^2 + 2pq + q^2 = 1$, told what p and q represent, shown how to calculate genotype frequencies from allele frequencies, and asked to reproduce the calculation in an examination. The equation arrived clean and complete, with its history removed, as if it had always existed in this form and had never been the product of a specific human mind working through a specific confusion on a specific afternoon.

The question the equation answers is this: if a trait is dominant, why does it not spread through a population over time until almost everyone shows the dominant phenotype? This question troubled serious scientists for years. It was asked publicly in 1908 by a trained quantitative biologist who had the mathematical tools to answer it and nevertheless could not see the answer. It was answered the same year by a mathematician who had never studied genetics and who thought the answer was so obvious that he was slightly embarrassed to be publishing it.

What made the difference between the person who could not see it and the person who could was not intelligence. It was the angle from which the problem was approached. This chapter is about that angle. It is also, underneath the algebra, about what it means to have a framework for answering a question that you have been circling without being able to answer, and how different the question looks once the framework arrives.

* * *

In 1908, a mathematics professor at Cambridge University solved a biology problem that had been troubling biologists for years. He solved it in an afternoon. He used arithmetic that most students learn before they are fifteen. And when he published the solution, he opened his paper by apologising for bothering anyone with something so obvious.

His name was G. H. Hardy. He was one of the finest pure mathematicians of his generation, and he was slightly embarrassed to have spent even an afternoon on a problem this simple. He spent the rest of his career insisting the result was trivial. He was wrong about that, but we will come to why in a moment.

The problem Hardy solved is one you have already encountered in your biology class, in the chapter on evolution. It is called the Hardy-Weinberg principle, and it appears in most class 12 biology textbooks as an equation to be memorised and applied in numerical problems. The equation is $p^2 + 2pq + q^2 = 1$. If you have seen this before, you probably learned what p and q represent and how to use the equation to calculate genotype frequencies. What you were probably not told is the story behind it, the confusion it dissolved, the reasoning that produced it, and why a pure mathematician who considered biology beneath his attention ended up writing one of the most widely used results in the history of population genetics.

That story is what this chapter is about.

* * *

To understand what Hardy did, you need to understand the confusion he was dissolving. And to understand the confusion, you need to understand a mistaken belief that many intelligent people held in the early twentieth century about how dominant traits behave in populations.

The mistaken belief was this: if a trait is dominant, it should spread through a population over time, because dominant traits mask recessive ones, and therefore organisms carrying dominant alleles have an advantage in expressing themselves phenotypically. The logical conclusion of this belief, if you followed it far enough, was that eventually all populations should consist almost entirely of individuals showing dominant phenotypes, with recessive phenotypes becoming increasingly rare until they more or less disappeared.

This belief was held not by ignorant people but by trained scientists who understood Mendelian genetics perfectly well at the level of individual crosses. They knew that when you cross two heterozygous parents, you get a three to one ratio of dominant to recessive phenotypes in the offspring. They extended this observation to whole populations and concluded that populations should over time tend toward a three to one ratio, with the dominant phenotype gradually taking over.

A biologist named Udny Yule made exactly this argument in public in 1908, using the example of brachydactyly, a condition that causes shortened fingers and is controlled by a dominant allele. If brachydactyly is dominant, Yule argued, then in the absence of any counteracting forces one would expect the proportion of brachydactylous people in the population to increase over time toward seventy-five percent. But this was clearly not happening. The proportion of brachydactylous people was remaining roughly constant at a much lower level. Yule presented this as a problem for Mendelian theory.

A geneticist named Reginald Punnett, whose name you know from the Punnett square, was present when Yule made this argument. Punnett knew Yule was wrong but could not immediately show exactly where the error was. When he got back to Cambridge he went to find his friend Hardy and explained the problem.

Hardy was not a biologist. He had no particular interest in genetics. What he had was a very clear understanding of probability, and when Punnett explained the problem to him, Hardy saw immediately that Yule had made a specific kind of error. Yule had taken a statement that is true at the level of one individual cross and applied it incorrectly to the level of an entire population across multiple generations. These are two completely different things, and confusing them produces a result that is mathematically wrong.

Hardy sat down and worked out what actually happens to allele frequencies in a population when you allow random mating across many generations. The calculation took him, by all accounts, a short time. The result he arrived at was that under certain conditions, allele frequencies in a population do not change from one generation to the next. They reach a stable distribution after just one generation of random mating and then remain there indefinitely. Dominant alleles do not increase in frequency. Recessive alleles do not decrease in frequency. The population sits in a kind of genetic equilibrium, generation after generation, unless something disturbs it.

This result demolished Yule's argument completely. Dominant traits do not spread through populations simply by virtue of being dominant. Dominance is a relationship between alleles inside a single organism. It has nothing to do with what happens to allele frequencies across a whole population over time. These are different levels of biological organisation, and what is true at one level is not automatically true at the other.

* * *

Now here is the question you should sit with for a moment before we look at the mathematics.

Why did it take biologists so long to see this? Hardy solved the problem in an afternoon using arithmetic that any mathematics student could follow. Punnett was an expert in Mendelian genetics. Yule was a trained quantitative scientist. Pearson, who led the biometrician group that Yule belonged to, was one of the founders of modern statistics. These were not intellectually limited people. Why did none of them see what Hardy saw so quickly?

The answer is not that Hardy was smarter than the biologists, though he was certainly a more accomplished mathematician. The answer is that Hardy was not looking at the problem as a genetics problem. He was looking at it as a probability problem. And seeing it as a probability problem rather than a genetics problem meant that the relevant question was immediately clear: what happens to the frequencies of different types of objects when you sample randomly from a population, combine them in pairs according to fixed rules, and then count the results in the next generation? This is a question about probability and combinatorics. It has a clean mathematical answer. Hardy could see the answer because he was not distracted by the biology.

This is one of the most important lessons in the history of scientific thinking, and it appears over and over again in the history of every scientific field. Sometimes the person who solves a problem is not the expert in that problem. Sometimes the expert is too close to the problem, too embedded in its particular vocabulary and its particular assumptions, to see the simple structure underneath. The outsider who knows less about the specific subject but more about a relevant general method can sometimes see what the expert cannot.

Hardy was that outsider in 1908. He knew almost nothing about genetics. He knew a great deal about how to think clearly about mathematical structures. And what population genetics needed at that moment was not more knowledge about genetics but clearer thinking about mathematical structure.

* * *

Let us now read what Hardy actually wrote. The paper is two pages long. It was published in the American journal Science in July 1908. Hardy published it there rather than in a British journal because British journals at the time were, as Punnett later explained, extremely hostile to anything associated with Mendelian genetics, which was then in the middle of a fierce dispute with the biometrician school led by Karl Pearson. That dispute, and what it tells us about how social and institutional forces shape scientific knowledge, is the subject of Chapter 2. For now, read the paper itself.

Read it slowly. Do not worry if some of the notation is unfamiliar. Read it as you would read a letter from someone who is trying to explain something they consider obvious but are willing to spell out because they have been asked. Notice the tone. Notice what Hardy chooses to say and what he chooses not to say. Notice the single sentence near the end where he gestures at something larger than the immediate problem without quite spelling it out. We will return to that sentence.

The paper appears on the following pages.

* * *

Now that you have read it, let us walk through what Hardy actually did, step by step, beginning with the algebra.

Hardy starts by imagining a population in which three genotypes exist: pure dominant, which we call AA, heterozygous, which we call Aa, and pure recessive, which we call aa. He says their proportions in the current generation are p, 2q, and r respectively. Notice that he uses 2q for the heterozygotes rather than simply q. This is because he is thinking of the alleles rather than the genotypes. If there are 2q heterozygotes, each carrying one A allele and one a allele, then the total count of A alleles in the heterozygote class is 2q, and the total count of a alleles is also 2q. This makes the subsequent algebra cleaner.

Now Hardy asks what happens in the next generation if mating is random. Random mating means that any individual in the population is equally likely to mate with any other individual, regardless of genotype. This is the first of several simplifying assumptions Hardy makes, and it is worth pausing on it because simplifying assumptions are not weaknesses in a mathematical model. They are choices. Every mathematical model of a real system ignores some features of that system in order to make progress on others. Hardy is choosing to ignore mate choice, geographic structure, variation in fertility, and many other real features of populations because he wants to isolate the question of what allele frequencies do under the simplest possible conditions. We will discuss what this choice means, and what it costs, after we have understood what the model shows.

Under random mating, the frequency with which any two alleles meet in a fertilised egg is simply the product of their individual frequencies in the population. The frequency of the A allele in the population is p plus q, because A alleles appear in both the pure dominant class and the heterozygote class. Call this frequency p plus q. Similarly, the frequency of the a allele is q plus r. Call this frequency q plus r.

Under random mating, the frequency of AA genotypes in the next generation is the probability that both alleles in a fertilised egg are A, which is the product of the allele frequencies: p plus q multiplied by p plus q, or p plus q squared. The frequency of aa genotypes is q plus r multiplied by q plus r, or q plus r squared. And the frequency of Aa heterozygotes is twice the product of the two allele frequencies, which is 2 multiplied by p plus q multiplied by q plus r. The factor of 2 appears because you can get a heterozygote in two ways, either by receiving A from the father and a from the mother, or by receiving a from the father and A from the mother.

So in the next generation the genotype frequencies are p plus q squared, 2 times p plus q times q plus r, and q plus r squared. Hardy now asks a simple question: under what conditions are these frequencies the same as the frequencies in the previous generation? That is, under what conditions does the population not change from one generation to the next?

The condition turns out to be q squared equals pr. If this condition holds in one generation, the algebra shows that it will hold in every subsequent generation, and the genotype frequencies will remain constant indefinitely. Hardy called this a stable distribution, always with inverted commas around the word stable, because he was precise enough as a mathematician to know that he was using the word in a weaker sense than its usual mathematical meaning. The distribution does not actively return to equilibrium if disturbed. It simply remains wherever it is, as long as nothing disturbs it.

The equation p squared plus 2pq plus q squared equals 1, which is the form you have likely seen in your textbook, is a simplified version of Hardy's result that uses a slightly different notation, where p and q represent the frequencies of the two alleles directly rather than Hardy's p, q, and r for the three genotype classes. The mathematics is equivalent. The form in the textbook is simply more compact.

* * *

There is something worth pausing on here, before we go further.

Hardy's model assumes random mating, a large population, no mutation, no migration, and no natural selection. Real populations violate all five of these assumptions simultaneously, all the time. Does that make the model useless?

No. It makes the model a baseline. This is one of the most important ideas in scientific modelling, and it is worth understanding clearly because it appears in different forms across every scientific discipline. A model that describes what would happen in the complete absence of any disturbing forces is not a description of reality. It is a reference point against which reality can be measured. When a real population deviates from Hardy-Weinberg equilibrium, that deviation is a signal that something is happening: selection, perhaps, or migration, or a population bottleneck, or non-random mating. The deviation does not tell you which of these things is happening. But it tells you that something is happening, and it gives you a quantitative measure of how much is happening, and that is often exactly what you need to know in order to start asking the right questions.

This is why Hardy-Weinberg equilibrium is still used in population genetics research more than a hundred years after Hardy wrote those two pages in Cambridge. Not because real populations are in equilibrium. They are not. But because the distance between a real population and the equilibrium prediction is one of the most informative measurements you can make.

Researchers studying earthworm populations in mining-affected landscapes in Odisha, or cheetah populations in southern Africa, or the genetics of isolated human communities in the Andes, all use Hardy-Weinberg equilibrium as a starting point. They measure allele frequencies. They calculate expected genotype frequencies under equilibrium. They compare expected to observed. The comparison tells them something about the evolutionary forces acting on that population in that place at that time.

Hardy did not know any of this when he wrote his two pages in 1908. He was answering a specific question that Punnett had brought to him, and he thought the answer was obvious, and he published it reluctantly. The uses to which that answer would be put across the following century were entirely invisible to him. This is another pattern that appears repeatedly in the history of science: the person who produces a piece of knowledge rarely knows what it will eventually be used for. Hardy's most useful result was a piece of work he considered too simple to be interesting. This should tell us something about how we evaluate the importance of scientific work at the moment it is produced.

* * *

Now we need to look at the sentence near the end of Hardy's paper that contains more than it announces.

After working through the algebra, Hardy writes that there is not the slightest foundation for the idea that a dominant character should show a tendency to spread over a whole population, or that a recessive should tend to die out.

This sentence is doing more than summarising the mathematical result. It is making a conceptual claim about the relationship between dominance as a molecular phenomenon and allele frequency dynamics as a population phenomenon. It is saying that these two things operate at different levels of biological organisation and that confusing them produces nonsense. Hardy makes this claim in a single sentence, without fanfare, at the end of a short paper, because to him it was the obvious implication of the obvious mathematics. But the claim is not obvious. It required someone to see the problem clearly enough to state it this cleanly. The mathematics made the clean statement possible. Without the mathematics, the conceptual point would have remained fuzzy, arguable, the kind of thing that intelligent people could disagree about indefinitely.

This is something you will see again and again as you read through the papers in this book. Mathematics does not just calculate things. At its best, it clarifies things. It forces a level of precision that dissolves confusions that would otherwise persist for years. Hardy's algebra did not discover that dominant traits do not spread through populations. In some sense, that fact was always there in the Mendelian rules themselves, waiting to be seen. What the algebra did was make it impossible to look away from.

* * *

In the next chapter we will read the paper that tells the story of how this result came to be named after Hardy and Weinberg, and what that story reveals about the social conditions that shape scientific recognition. But before you go there, spend a few minutes with the questions below. Not because you will be examined on them. Because sitting with a question long enough to form your own view about it is what reading this book is for.

Why did Hardy use inverted commas around the word stable? What was he worried about? What does stability mean in mathematics, and why might the Hardy-Weinberg distribution not quite meet that standard?

Hardy assumed random mating. Think about a real population you know, whether human, animal, or plant, in your own environment. In what ways does mating in that population depart from randomness? What would those departures do to allele frequencies over time?

Hardy thought his result was trivial. A century later it is one of the most widely applied results in biology. What does this tell you about how we should evaluate the importance of a scientific result at the moment it is produced?

* * *

Hardy's paper answers the question it was asked. Dominant traits do not spread through populations simply by virtue of being dominant, because allele frequencies in a large randomly mating population reach a stable distribution after one generation of random mating and then remain there unless something disturbs them. The mathematics is clear. The conclusion is decisive.

But the paper raises a question it does not answer, and it is a question that turns out to be as important as the result itself. If the argument is this simple, if it requires only arithmetic and a clear statement of the problem, why did it take until 1908 for anyone to make it? Yule had the relevant mathematical tools. Pearson had stronger mathematical tools than Hardy himself. Punnett had the biological knowledge. The components of the argument were all available, in the hands of people who were actively working on related problems, for at least six years before Hardy wrote his two pages. Why did nobody see it?

The next chapter is about that question. It is the question Edwards spent an entire paper trying to answer, and the answer he arrived at reveals something about scientific discovery that the discovery itself could not reveal. A result shows you what was found. The history of how the result was found shows you what scientific thinking actually is, which turns out to be rather different from what the result alone would lead you to believe.

Chapter 2

Why It Took Six Years to See Something Obvious

A. W. F. Edwards, 2008

There is a particular kind of historical puzzle that appears regularly in the history of science, and it goes like this: a result that seems straightforward in retrospect, one that requires no exotic equipment, no unusual mathematical machinery, no data that was unavailable at the time, nevertheless goes unseen for years while intelligent and well-trained people work on the surrounding problem without quite arriving at it. When you eventually trace the history carefully, you find that the result was almost seen several times by people who had everything they needed to see it. They had the right mathematical tools. They had the right biological knowledge. They had, in some cases, even written down pieces of the argument. But they stopped just short of putting the pieces together, and someone else, often someone who was not even trying to solve that particular problem, put them together later.

The Hardy-Weinberg principle is one of the cleaner examples of this pattern in the history of biology. Hardy published his two-page paper in July 1908. Weinberg published his independent derivation in January of the same year. But the logical components of the result had been available since at least 1902, and the people who almost saw it first were not marginal figures working at the edges of the relevant fields. They were Udny Yule, who was a sophisticated quantitative biologist with strong mathematical training, and Karl Pearson, who was one of the founders of modern statistics, a man who had essentially invented several of the analytical tools that population genetics would later depend on. Both of them wrote down arguments that came very close to the Hardy-Weinberg result without quite arriving at it. Both of them stopped at a point where, with one more step, they would have been there.

Why? That is the question Edwards' paper is trying to answer, and it is a more interesting question than it might initially appear, because the answer is not simply that Yule and Pearson were not smart enough or not trying hard enough. The answer involves the structure of scientific communities, the effect of intellectual disputes on the ability to think clearly, and the way that the language in which you frame a problem shapes which solutions you can see.

This chapter is about that answer. But it is also about something larger, which is what it means to read a paper about the history of a scientific idea rather than a paper reporting a scientific discovery. The five papers we will examine in the other chapters of this book are primary papers, meaning they are the original reports of the work itself. This chapter's paper, written by the mathematician and historian A. W. F. Edwards and published in the journal Genetics in 2008, exactly one hundred years after Hardy's original letter, is a secondary paper, meaning it is a paper about a paper. It asks not what Hardy discovered but how and why Hardy came to discover it when he did, and what the circumstances of the discovery tell us about the social conditions in which scientific knowledge is produced.

Reading a secondary paper of this kind requires a different kind of attention than reading a primary paper. In a primary paper, you are asking: what is being claimed, what evidence supports the claim, and what kind of reasoning connects the evidence to the claim? In a secondary paper about the history of science, you are asking: what story is being told, what sources does the story rest on, what alternative stories might be told about the same events, and what does this particular story reveal about the way scientific communities work that would not be visible if you looked only at the primary papers themselves?

Both kinds of reading are essential. If you read only primary papers, you see the finished products of scientific thinking but not the human processes that produced them. If you read only historical analyses, you understand the processes but lose sight of the actual scientific content. Reading them together, as this book asks you to do, gives you something that neither alone can provide: a view of science as it actually is, which is a human activity conducted by specific people in specific places under specific institutional and social conditions, producing knowledge that is real and reliable but whose production is far messier and more contingent than any textbook account suggests.

* * *

To understand what Edwards is explaining, you need to know something about the intellectual landscape of British biology in the first decade of the twentieth century, because that landscape was dominated by a dispute that was, even by the standards of academic disputes, unusually bitter and unusually consequential.

On one side were the biometricians, led by Karl Pearson at University College London. Pearson and his colleagues believed that heredity should be studied statistically, by measuring traits across large numbers of individuals and analysing the patterns in those measurements. They were deeply sceptical of Mendel's particulate theory of inheritance, which proposed that traits were controlled by discrete factors, what we now call genes, that were passed from parent to offspring in specific ratios. The biometricians thought this picture was too simple, that it failed to account for the continuous variation they observed in most traits in natural populations, and that it was being promoted by people who were more interested in scoring points against the statistical approach than in actually understanding heredity.

On the other side were the Mendelians, led by William Bateson at Cambridge. Bateson and his colleagues, including Punnett, were convinced that Mendel's framework was correct and that it provided the key to understanding inheritance at the level of discrete hereditary units. They thought the biometricians were measuring the outcomes of hereditary processes without understanding the mechanisms underneath, and they were not always tactful in saying so.

The dispute between these two groups was not merely scientific. It was territorial, personal, and conducted in a tone that made genuine intellectual exchange extremely difficult. Pearson's journal Biometrika effectively refused to publish Mendelian work. Bateson's group responded in kind. The two communities had largely stopped reading each other's papers carefully by the time Hardy entered the picture in 1908, which meant that results produced on one side of the divide were often invisible to people on the other side, even when those results were directly relevant to questions being asked on both sides.

This is the first part of the answer to why it took so long. The communities were not talking to each other. Yule and Pearson, working within the biometrician tradition, had the mathematical tools to derive the Hardy-Weinberg result. Punnett and Bateson, working within the Mendelian tradition, had the biological knowledge to understand why the result mattered. But the two traditions were not exchanging information effectively, and so the person who had the mathematical tools was not asking the right biological question, and the people who were asking the right biological question did not have the mathematical clarity to answer it.

The second part of the answer is more subtle. Yule and Pearson were not just missing the biological context. They were trapped by a specific assumption about gene frequencies that seems, in retrospect, to have been so obvious to them that they never questioned it explicitly. They appear to have assumed, without quite stating the assumption, that the natural baseline for a Mendelian population was a fifty-fifty ratio of dominant to recessive alleles, the ratio you get when you start with an equal number of AA and aa parents and allow them to cross. If you make that assumption, then the Hardy-Weinberg result follows almost trivially, and you see only the special case, not the general principle. Neither Yule nor Pearson stepped back far enough to ask what happens when the allele frequencies are not equal, which is of course the general case and the one that matters for real populations.

It is worth pausing on what Weinberg's story actually shows, because the distance of a hundred years can make it feel like a resolved problem rather than a continuing one.

Weinberg published in German in a regional journal. Hardy published in English in an international one. The science was the same. The reach was not. And the reach determined, for many years, whose name attached to the result. This was not the outcome of anyone's deliberate choice. It was the outcome of where each man sat within the networks through which scientific knowledge travels, which journals circulate widely, which languages carry ideas across borders, which institutional addresses signal to other scientists that the work coming from them is worth reading carefully.

Those networks have changed since 1908. They have not been replaced by something neutral. A scientist working today from an institution outside the major centres of global science, publishing work that does not reach the journals with the widest circulation, still faces a version of the same asymmetry that Weinberg faced. The asymmetry is smaller. It has not disappeared.

This is mentioned here not as grievance but as context. Understanding how scientific recognition actually works, rather than how we imagine it works, is part of understanding science. Edwards' paper does that understanding the service of being precise about it rather than polite about it. That precision is part of what makes it worth reading carefully.

* * *

Hardy stepped back that far because he was not thinking about genetics. He was thinking about a probability problem with arbitrary initial frequencies, and arbitrary initial frequencies are the natural starting point for a mathematician approaching the problem fresh, without the preconceptions that years of work in a specific field accumulate.

* * *

There is a third part of the answer, and it is the one that Edwards dwells on most carefully, perhaps because it is the most uncomfortable. Yule, in his 1902 paper, actually derived the Hardy-Weinberg result for the special case of equal allele frequencies. He wrote it down. He then moved on without recognising what he had written, because he was interested in a different question at that moment and the result he had derived was not the answer to the question he was asking. He had the answer in his hands and did not see it, because he was not looking for it.

This is not an unusual experience in science. The history of science is full of people who had important results in their notebooks or their published papers and did not recognise them as such, because recognition requires not just calculation but the understanding of what a calculation means, and that understanding depends on asking the right question. Yule was asking about the regression of offspring on parents. The Hardy-Weinberg equilibrium was not the answer to that question. It was the answer to a different question, one that Punnett eventually brought to Hardy in the right form for Hardy to see it clearly.

The particular form in which Punnett brought the question to Hardy matters here. Punnett had interpreted Yule's objection, somewhat imprecisely, as the question of why dominant genes did not increase in frequency over time, rather than the more specific question of why populations did not converge to a three to one ratio. These are related questions but they are not identical, and the slightly different framing that Hardy received is, Edwards argues, part of what made Hardy's answer take the form it did. Hardy answered the question he was asked, which was the question about gene frequencies under random mating, and in answering it he produced the general result. If Punnett had framed the question more narrowly, Hardy might have produced a narrower answer.

This is another pattern that appears repeatedly in the history of scientific discovery: the precise form in which a question is asked shapes the form of the answer that becomes possible. Scientists who are asked narrow questions tend to produce narrow answers. Scientists who are asked broad questions, or who reframe a narrow question into a broader one before answering it, tend to produce more general results. The ability to reframe a question is one of the most important intellectual skills in science, and it is almost never taught explicitly, in textbooks or classrooms, because it is difficult to reduce to a rule or a method. It is more like a habit of mind than a technique, and it develops through practice and through reading accounts like Edwards' paper that make the habit visible.

* * *

Now let us turn to what Edwards' paper reveals about the question of credit and recognition.

Wilhelm Weinberg was a physician in Stuttgart who derived the Hardy-Weinberg result independently in January 1908, four months before Hardy published his letter in Science. Weinberg published his result in German in a regional German natural history journal. He went further than Hardy in some respects, working out the implications of the principle for multiple loci and developing methods to use it for the analysis of human hereditary data. His contribution was, by any reasonable assessment, at least as significant as Hardy's and in some respects more so.

For many years after 1908, the principle was known simply as Hardy's law, without any mention of Weinberg. Weinberg's contribution became widely recognised in the English-speaking world only decades later, largely because his paper was published in German at a time when German-language science was becoming increasingly inaccessible to English-speaking scientists, and partly because Weinberg was working outside the academic network of British and American universities where the principle was being applied and taught. He was a practising physician, not a university professor. He did not have students who would carry his name into the literature. He did not have colleagues at prestigious institutions who would cite him in their textbooks.

Hardy, by contrast, was at Cambridge. His result was communicated to the relevant biological community immediately by Punnett, who was also at Cambridge and who was deeply embedded in the Mendelian network. The institutional location of the person who produces a result shapes how quickly and how completely that result becomes part of the scientific conversation, and this is not a trivial or accidental feature of how science works. It is a systematic feature, one that operates to amplify the recognition of results produced at prestigious institutions in major scientific languages while attenuating the recognition of results produced elsewhere, regardless of their scientific merit.

For a student reading this book in India, or in Nigeria, or in Brazil, or in any country whose scientific institutions were built after independence and whose scientists work in conditions that are less well-resourced and less well-networked than those of the major scientific centres in Europe and North America, this observation is not merely historically interesting. It is directly relevant to the conditions under which you and your teachers are working right now. The mechanisms that delayed Weinberg's recognition in 1908 are still operating today. They operate more slowly and less decisively than they did then, because the scientific community is larger and more geographically distributed and because English is now genuinely the international language of science in a way it was not in 1908. But they operate.

This is not said to discourage anyone. It is said because understanding the conditions under which science is produced and recognised is part of scientific literacy, and pretending those conditions are neutral when they are not is a form of dishonesty that serves no one well.

* * *

There is one more thing Edwards' paper does that is worth drawing attention to before you read it.

Edwards shows, with considerable care, that the priority question, the question of who deserves credit for the Hardy-Weinberg principle, does not have a simple answer, because the question of what exactly the principle is depends on how precisely you define it. Yule had a version of it in 1902. Pearson had a version of it in 1904. Castle, an American geneticist, had a numerical version of it in 1903. Weinberg had the general version in January 1908. Hardy had the general version in July 1908. Each of these people saw something, and what each of them saw was slightly different from what the others saw.

Scientific priority, in other words, is not usually a matter of one person discovering something that was previously entirely unknown. It is more often a matter of one person stating something with sufficient generality and clarity that the scientific community recognises it as a principle worth naming and using. Hardy's contribution was not that he was the first to see that allele frequencies stabilise under random mating. His contribution was that he stated this clearly enough, in a sufficiently general form, and in a sufficiently prominent venue, that the result became available to the wider scientific community in a usable form.

Clarity and generality and accessibility, in other words, are not just stylistic virtues in scientific writing. They are scientific contributions in their own right. The person who states something clearly enough to be understood and used by others has done something scientifically valuable, even if someone else saw the same thing first but stated it in a form that the community could not pick up and use. This is a principle worth carrying with you as you read the remaining papers in this book, and as you eventually begin to write your own scientific work.

* * *

Now read Edwards' paper. It is longer and more discursive than Hardy's paper, because it is doing a different kind of work. Hardy was making an argument in algebra. Edwards is making an argument in prose, reconstructing a historical episode from primary sources, including the published papers of the participants, archival records, and the recollections of Punnett himself, recorded in a lecture he gave in 1950. Read it as you would read any careful historical argument: asking not just what happened but why, and what the account reveals about the conditions that made it possible for things to happen the way they did rather than some other way.

After you have read it, here are the questions worth sitting with.

Yule had the Hardy-Weinberg result in 1902 but did not recognise it as such. Has something like this ever happened to you, in your own thinking? Have you ever had an answer to a question you were not asking, and not recognised it until later?

What does this suggest about the relationship between questions and the ability to see answers?

Weinberg published in German in a regional journal. Hardy published in English in an international journal. The result was the same. The recognition was very different. What would need to change, in how scientific communities are organised, for this kind of disparity to be reduced?

Edwards argues that clarity and generality in stating a result are scientific contributions in their own right, not just stylistic improvements. Do you agree? Can you think of an example, from your own experience of learning science, where a clearer or more general statement of something you already knew changed what you could do with that knowledge?

* * *

Edwards' paper shows us that scientific knowledge is produced within social and institutional conditions that shape what gets seen, by whom, and when. Hardy saw what Yule and Pearson did not see partly because Hardy was an outsider who brought a mathematician's clarity to a problem that biologists had framed in a way that obscured its structure. The institutional feud between the biometricians and the Mendelians meant that the people who had the mathematical tools and the people who had the biological knowledge were not talking to each other effectively enough to combine what they knew.

This is one kind of obstacle to scientific discovery: the social and institutional obstacle, the feud, the disciplinary boundary, the geographical distance, the language of publication. But there is another kind of obstacle that is more fundamental, and it is the kind that the next paper encountered in its most dramatic form. Sometimes the obstacle is not social but physical. The thing you want to know is written in matter, in the structure of molecules, and the question is whether you can read it from the outside, from the patterns it casts when bombarded with X-rays, from the constraints its chemistry imposes, from the geometry of the space it must occupy.

Watson and Crick were trying to read a physical structure from indirect evidence, and the paper they wrote is the record of how they did it. It is also, in that last quiet sentence, the record of what they understood they had found when they found it.

Chapter 3

Nine Hundred Words and One Quiet Sentence

Watson and Crick, 1953

On the twenty-fifth of April 1953, the journal Nature published a paper that was less than a thousand words long, contained one figure, cited only six previous publications, and changed biology more completely than any other paper published in the twentieth century. The paper was written by James Watson, a twenty-four year old American postdoctoral researcher, and Francis Crick, a thirty-six year old British physicist who had not yet completed his doctorate. Neither of them had solved the problem they were publishing about by doing experiments in the conventional sense. They had built a physical model out of metal rods and wire and flat pieces of cardboard cut into the shapes of the relevant molecules, and they had argued, from the geometry of that model and from published experimental data gathered mostly by other people, that they had found the correct structure of deoxyribose nucleic acid.

What makes this paper worth reading slowly, rather than reading for its conclusions, is that its most important claim is not in its conclusions. It is in a subordinate clause near the end, written in deliberately cautious language, pointing toward something the authors could see but could not yet prove. Missing that sentence, which is easy to do if you are reading quickly, means missing the most important thing the paper does. This chapter is an exercise in reading slowly enough to find it.

* * *

The structure they proposed was the double helix. You have seen diagrams of it in your textbook. You have probably been told that it consists of two polynucleotide chains wound around each other in a right-handed spiral, with the sugar-phosphate backbones on the outside and the nitrogenous bases pointing inward, paired across the middle of the structure according to specific rules: adenine with thymine, guanine with cytosine. You have probably memorised these base pairing rules and been asked to apply them in examinations. What you have probably not been told is why the paper that proposed this structure was revolutionary in a way that goes far beyond the structure itself, or why the most important thing in the paper is not the structure but a single sentence near the end that Watson and Crick wrote with deliberate understatement, almost as if they were mentioning it in passing.

That sentence is what this chapter is about. But to understand why the sentence matters, you first need to understand what problem Watson and Crick were solving and why it had proved so difficult.

* * *

By 1953, biologists had known for nearly a decade that DNA was the genetic material. The experiments of Avery, MacLeod and McCarty in 1944, confirmed by Hershey and Chase in 1952, had established beyond reasonable doubt that it was DNA, not protein, that carried hereditary information from one generation to the next. This was, in itself, a surprising finding, because DNA had seemed to many biologists too simple a molecule to carry all the information needed to specify a living organism. DNA contains only four types of bases, adenine, thymine, guanine, and cytosine, while proteins contain twenty types of amino acids, and it had seemed natural to assume that the more chemically complex molecule was the one carrying the more informationally complex message.

But once it was established that DNA was the genetic material, a new and urgent question arose. How does DNA carry information? And more pressingly, how does DNA copy itself accurately enough that the information it carries can be transmitted reliably from one cell to its daughter cells, and from one generation of organisms to the next? These were not the same question, but they were related, and any proposed structure for DNA had to be consistent with plausible answers to both of them.

Several groups were working on the structure of DNA in the early 1950s. Rosalind Franklin and Maurice Wilkins at King's College London were using X-ray crystallography to produce diffraction patterns from DNA fibres, and their data was providing the most detailed experimental information available about the dimensions and geometry of the molecule. Linus Pauling and Robert Corey at Caltech had recently proposed a triple-helical structure for DNA that turned out to be incorrect, partly because Pauling had not had access to the most recent X-ray data from King's College. Watson and Crick at Cambridge were building models, trying to find a structure that was consistent with the known chemistry of the bases, the known geometry of the phosphate backbone, and the X-ray data.

The X-ray data that proved most important to Watson and Crick was a photograph taken by Rosalind Franklin in 1952, known as Photo 51, which showed an unmistakably helical diffraction pattern and provided precise measurements of the repeat distances in the helix. Watson saw this photograph, without Franklin's knowledge or permission, when Wilkins showed it to him during a visit to King's College in January 1953. The photograph told Watson immediately that the structure was helical and gave him the key measurements he needed to build a model that fit the data. Within weeks, Watson and Crick had built the double helix model and written the paper.

The question of credit in this story is one that biology has been grappling with ever since. Franklin's experimental data was essential to the solution. She was not told that her photograph had been shown to Watson, and she was not consulted about the model Watson and Crick were building. She died of cancer in 1958, four years before Watson, Crick, and Wilkins received the Nobel Prize for the discovery of the structure of DNA. The Nobel Prize is not awarded posthumously, so Franklin could not have shared in it even if the committee had wanted to include her, but the question of whether she was given adequate credit during her lifetime, and whether the use of her data without her knowledge was ethically defensible, has remained a live and uncomfortable one in the history of molecular biology.

This context matters not because it diminishes what Watson and Crick achieved, which was genuinely extraordinary, but because it is part of the complete story of how the discovery was made, and incomplete stories about scientific discovery create false impressions about how science works. Science is not conducted by isolated geniuses who derive results purely from their own intellect. It is conducted by people who build on each other's work, who operate within networks of collaboration and competition, and whose access to each other's data is governed by norms that are not always followed and that are sometimes, as in this case, actively violated. Knowing this does not make the science less impressive. It makes the account of the science more honest.

* * *

With that context in place, read the paper. It is short enough to read twice, and you should read it twice. On the first reading, simply follow the argument: what structure is being proposed, what evidence is cited in support of it, what alternative structures are being rejected and why. On the second reading, pay attention to the language. Notice how Watson and Crick describe the relationship between their structure and the experimental data. Notice which claims are made with confidence and which are hedged. Notice the sentence near the end that begins with the phrase it has not escaped our notice.

Read it now, before continuing.

* * *

The structure Watson and Crick proposed has several features that are worth understanding carefully, because each of them does specific biological work.

The two chains run in opposite directions. If you follow one chain from the end where the phosphate group is attached to the 5' carbon of the sugar, you are moving in what biochemists call the 5' to 3' direction. The other chain runs in the opposite direction, from 5' to 3' in the other direction. This antiparallel orientation, as it is called, is not arbitrary. It is a consequence of the chemistry of the phosphodiester bonds that link nucleotides together, and it has profound consequences for the way DNA is replicated and transcribed.

The bases pair in a specific and non-arbitrary way. Adenine pairs with thymine, and guanine pairs with cytosine. This specificity is a consequence of the precise geometry of hydrogen bonding between the bases. An adenine base has a shape and a pattern of hydrogen bond donors and acceptors that fits precisely with the shape and hydrogen bonding pattern of thymine, and the same is true for guanine and cytosine. Any other pairing would either produce too many clashing atoms or too few hydrogen bonds to hold the structure together. The base pairing rules are therefore not rules that biologists decided to impose on the molecule. They are rules that the geometry of the molecule imposes on itself.

This geometric specificity of base pairing is what makes the double helix capable of carrying information. A sequence of bases along one chain of the helix determines, with complete precision, what the sequence of bases on the other chain must be. If one chain reads ATGCCTA in the 5' to 3' direction, the complementary chain must read TACGGAT in the 5' to 3' direction, because no other sequence is geometrically possible given the pairing rules. The two chains are therefore not identical but complementary, and this complementarity is the key to understanding everything that the double helix makes possible.

* * *

Now we come to the sentence.

Near the end of the paper, after Watson and Crick have described the structure and noted that it is consistent with the available X-ray data, they write the following: it has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material.

Read that sentence again slowly.

What Watson and Crick are pointing to, in this single quietly written sentence, is something that had never been true of any proposed biological molecule before. They are saying that the structure of DNA, specifically the complementary base pairing between the two strands, implies a mechanism by which the molecule could copy itself. If you separate the two strands of the double helix and allow each strand to serve as a template for the synthesis of a new complementary strand, you would end up with two double helices, each identical to the original. The information carried in the sequence of bases would be preserved perfectly, in both copies, because the base pairing rules constrain each new strand to be the exact complement of its template, and the complement of the complement is the original sequence.

This is the copying mechanism that Watson and Crick are suggesting with their understated phrase. And what makes the suggestion so important is not just that it provides an answer to the question of how DNA copies itself, though it does provide that answer, and the answer turned out to be correct, as Meselson and Stahl demonstrated five years later in the experiment we will examine in the next chapter. What makes it important is the relationship between the structure and the mechanism. The structure does not merely allow a copying mechanism. It makes a copying mechanism almost inevitable. You cannot have a double helix with complementary base pairing without having, built into the very geometry of the molecule, a template for its own duplication.

This had never been true before of any proposed genetic material. The relationship between the structure of a molecule and its biological function had always been something that required additional explanation, additional mechanisms, additional components. Here, for the first time, the structure explained the function directly. The shape of the thing was also the mechanism by which the thing did what it needed to do. Form and function were not just related but identical.

Watson and Crick understood the magnitude of what they were saying. Crick reportedly told everyone in the pub that evening that they had found the secret of life, which was not a significant exaggeration. But in the paper, they said it in one sentence, in a subordinate clause, using the word suggests rather than proves and the word possible rather than certain, because they did not yet have experimental proof that DNA actually replicated by this mechanism. They had a structure that implied a mechanism. The mechanism still needed to be demonstrated.

The restraint in that sentence is not false modesty. It is scientific precision. Watson and Crick knew exactly what they had found and exactly what they had not yet proven, and the language of the sentence reflects that distinction perfectly. This kind of precision in the use of language is one of the things that separates scientific writing from other kinds of writing, and one of the things that makes reading scientific papers a different kind of intellectual activity than reading other kinds of texts. Every word in a scientific claim carries weight. The difference between suggests and demonstrates, between possible and actual, between consistent with and proved by, is not a stylistic difference. It is a difference in what the author is claiming to know and with what degree of certainty.

* * *

There is one more thing to notice about the paper before we move to the broader questions it raises.

Watson and Crick, in their opening paragraphs, explicitly discuss and dismiss two alternative structures for DNA that had been proposed by other groups: the triple-helical structure of Pauling and Corey, and an alternative helical structure proposed by Fraser. They give reasons for rejecting each. This is standard practice in scientific papers, but it is worth paying attention to what this practice reveals about how scientific arguments are structured. A scientific paper does not just present a positive claim and the evidence for it. It also addresses the alternatives, explaining why the alternatives are less consistent with the available evidence than the proposed solution. A claim that is not tested against alternatives is weaker than a claim that survives the comparison.

The specific objections Watson and Crick raise against Pauling's triple helix are worth reading carefully. They point out that in Pauling's structure, the phosphate groups are near the centre of the helix rather than on the outside, and that this requires the phosphates to be unnegatively charged, which is chemically implausible under the conditions in which DNA normally exists in cells. This is a chemistry argument, not a mathematical or a geometric one, and it illustrates the fact that even a paper that is primarily about molecular structure draws on multiple kinds of knowledge simultaneously. The evidence for the double helix came from X-ray crystallography, which is physics. The argument against the triple helix rested on chemistry. The significance of the base pairing rules required biology. Good science rarely stays within disciplinary boundaries, and one of the things you will notice as you read more primary papers is how often the most important steps involve importing a tool or a piece of knowledge from one field into another.

* * *

The questions this chapter leaves you with are these.

Watson and Crick write that their structure suggests a possible copying mechanism rather than stating directly that DNA copies itself by strand separation and template synthesis. Why does this caution matter? What would have been lost if they had stated the mechanism as a fact rather than a suggestion?

Rosalind Franklin's data was essential to the discovery of the double helix. She was not adequately credited during her lifetime. Does the scientific validity of Watson and Crick's result depend on whether the process by which they obtained Franklin's data was ethically defensible? Is scientific truth separable from the ethical conditions under which it is discovered?

The double helix makes its own copying mechanism geometrically inevitable. Can you think of another example, from any field, where the structure of something implies its function so directly that knowing the structure is equivalent to knowing the mechanism? What does it mean for form and function to be identical rather than merely related?

* * *

Watson and Crick's paper ends with a suggestion. The specific pairing they have postulated, they write, immediately suggests a possible copying mechanism for the genetic material. They write this in one sentence, in a subordinate clause, using the word suggests rather than demonstrates and the word possible rather than actual. They are pointing toward something they cannot yet prove.

The distance between suggests and demonstrates is where experimental science lives. Watson and Crick could see, from the geometry of the double helix, that the copying mechanism was implied by the structure. But seeing an implication is not the same as demonstrating that the implication describes what actually happens. To demonstrate that, you need an experiment, and the experiment you need is not just any experiment that is consistent with the proposed mechanism. You need an experiment whose design is so precise, so carefully matched to the logical structure of the question, that the result speaks unambiguously. An experiment that can be interpreted in more than one way is not a demonstration. It is a suggestion made in a different language.

The next chapter is about the experiment that turned Watson and Crick's suggestion into a demonstration. It is also about what makes an experiment beautiful, which turns out to be a more precise and more useful concept than it might initially appear.

Chapter 4

The Most Beautiful Experiment in Biology

Meselson and Stahl, 1958

In the summer of 1957, two young scientists at the California Institute of Technology set themselves the task of designing an experiment that would settle, once and for all, a question that the Watson-Crick paper had left unanswered. The question was this: when a DNA molecule copies itself, what physically happens to the two original strands?

Watson and Crick had suggested, in that one quiet sentence at the end of their 1953 paper, that the copying mechanism probably involved the separation of the two strands, with each strand serving as a template for the synthesis of a new complementary strand. But suggesting a mechanism and demonstrating it are different things, and in 1957 the mechanism had not been demonstrated. Three distinct possibilities were still on the table, each of them consistent with what was known about DNA chemistry at the time, and each of them making different predictions about what would happen to the original strands during copying.

The first possibility was the one Watson and Crick had suggested, which is called semiconservative replication. In semiconservative replication, the two original strands separate completely, and each serves as a template for a new complementary strand. The result is two daughter DNA molecules, each containing one original strand and one newly synthesised strand. The original molecule is not conserved as a unit. Its strands are distributed between the two daughter molecules.

The second possibility is called conservative replication. In conservative replication, the original double helix somehow serves as a template for the synthesis of an entirely new double helix, while remaining intact itself. The result is two daughter molecules, one of which is the original molecule, completely unchanged, and one of which is entirely new. The original molecule is conserved as a unit.

The third possibility is called dispersive replication. In dispersive replication, the original strands are broken into fragments during copying, and the new and old material is distributed randomly among the fragments of the daughter molecules. The result is two daughter molecules, each containing a mixture of old and new material distributed throughout their length, with no intact segments of original DNA preserved anywhere.

These three possibilities were not just theoretical curiosities. They reflected genuinely different ideas about the physical mechanism of DNA replication, and distinguishing between them experimentally required measuring something that had never been measured before: the distribution of old and new material within individual DNA molecules after replication.

The two scientists who took on this problem were Matthew Meselson, a twenty-six year old graduate student in chemistry, and Franklin Stahl, a postdoctoral researcher in biology. They were an unlikely pair in some respects, working across disciplinary lines in a way that was not common at the time, and their collaboration had begun somewhat informally, over conversations in the cafeteria at Caltech and during a long drive from Colorado to California after a summer meeting. The experimental strategy they eventually developed was elegant in a way that is worth understanding carefully, because its elegance is not accidental. It is the result of a specific kind of thinking about experimental design that is as important to learn as any particular piece of biological knowledge.

* * *

The central challenge Meselson and Stahl faced was this: old DNA and new DNA, synthesised from the same four bases, are chemically identical. You cannot tell them apart by any ordinary chemical test. If you cannot distinguish old DNA from new DNA, you cannot measure how the old and new material is distributed in the daughter molecules after replication, and if you cannot measure that distribution, you cannot distinguish between the three proposed mechanisms.

The solution Meselson and Stahl arrived at was to make old DNA and new DNA physically distinguishable, not by changing their chemistry but by changing their mass. They did this by growing bacteria, specifically Escherichia coli, for many generations in a growth medium containing a heavy isotope of nitrogen, nitrogen-15, instead of the normal light isotope, nitrogen-14. Nitrogen is a component of all four DNA bases, and bacteria grown in nitrogen-15 medium incorporate nitrogen-15 into their DNA rather than nitrogen-14. After many generations of growth in this heavy medium, essentially all the nitrogen in the bacterial DNA is nitrogen-15, and the DNA is measurably heavier than normal DNA containing nitrogen-14.

Meselson and Stahl then transferred the bacteria to normal growth medium containing nitrogen-14 and allowed them to replicate their DNA once, twice, and several more times. Any DNA synthesised after the transfer would contain nitrogen-14 and would therefore be lighter than the original nitrogen-15 DNA. The question was how the heavy original DNA and the light newly synthesised DNA would be distributed in the daughter molecules.

To answer this question, they needed a way to separate DNA molecules of different masses from each other. The technique they used, which Meselson had developed in collaboration with Jerome Vinograd, was density gradient centrifugation in a solution of caesium chloride. When a caesium chloride solution is spun at very high speed in an ultracentrifuge for many hours, the caesium chloride molecules distribute themselves along a density gradient, with higher concentrations near the bottom of the tube and lower concentrations near the top. DNA molecules suspended in this solution migrate to the position in the gradient where their density exactly matches the density of the surrounding caesium chloride solution. DNA molecules of different densities migrate to different positions. If you then shine ultraviolet light through the tube and photograph it, the DNA appears as a dark band at its equilibrium position. Different populations of DNA molecules with different densities appear as different bands at different positions in the tube.

This technique was, at the time, genuinely new. Meselson and Stahl were not just applying an existing tool to a new problem. They were developing a tool specifically for this experiment, which is itself a lesson worth noting. The most important experiments in the history of science have often required the development of new experimental methods, not just the application of existing ones. The question comes first, and the method for answering it is developed afterward, shaped by the specific demands of the question. Meselson and Stahl needed to separate DNA molecules by density, and they developed a way to do it.

* * *

The results of the experiment were remarkably clean, and their cleanliness is a large part of what has made this experiment famous.

When Meselson and Stahl extracted DNA from bacteria that had been grown entirely in nitrogen-15 medium, and subjected it to density gradient centrifugation, it formed a single band at a position corresponding to heavy DNA, as expected. When they extracted DNA from bacteria after one generation of growth in nitrogen-14 medium, the DNA formed a single band, but at a position intermediate between heavy and light DNA. There was no heavy band, and no light band. Just one band, in the middle.

This single result ruled out conservative replication immediately and completely. Conservative replication predicts that after one generation, you should see two bands: one heavy band from the original DNA molecules, which are conserved intact, and one light band from the entirely new DNA molecules. The experiment showed only one intermediate band. Conservative replication was wrong.

The intermediate position of that single band was exactly what semiconservative replication predicts. If each daughter molecule contains one original heavy strand and one newly synthesised light strand, the average density of each molecule will be exactly halfway between heavy and light, and all daughter molecules will have the same intermediate density, producing a single band at the midpoint.

After two generations of growth in nitrogen-14 medium, the DNA formed two bands: one at the intermediate position and one at the light position. Again, this is exactly what semiconservative replication predicts. After two generations, half the molecules should contain one original heavy strand and one new light strand, giving intermediate density, and half should contain two new light strands, giving light density. The dispersive replication model, which predicts that after two generations the old and new material should be randomly intermixed throughout all molecules, would produce a single band somewhere between intermediate and light, not two distinct bands. The two-band result ruled out dispersive replication.

In two generations of bacteria and a handful of centrifuge tubes, Meselson and Stahl had distinguished between three competing models, ruled out two of them, and confirmed the third. The experiment was complete.

* * *

The physicist Max Delbruck, who was one of the founders of molecular biology and who had himself proposed a version of dispersive replication that he was rather attached to, was reportedly converted immediately upon seeing the results. The immunologist and Nobel laureate Peter Medawar later called it the most beautiful experiment in biology. It is worth thinking carefully about what Medawar meant by beautiful, because beautiful is not a word that is usually applied to scientific experiments, and understanding why it applies here tells you something important about experimental design that goes beyond this particular experiment.

The beauty of the Meselson-Stahl experiment lies not in the visual appearance of the results, though the autoradiograph images of the bands in the centrifuge tubes are striking in their clarity. It lies in the relationship between the design of the experiment and the structure of the question being asked. The three competing models each make different and distinguishable predictions about what the density distribution of DNA should look like after one generation and after two generations of replication in light medium. The experimental design was constructed so that those predictions could be tested directly, with results that were unambiguous. The question had exactly three possible answers, the experiment was designed so that each answer would produce a different observable outcome, and the results pointed unambiguously to one of the three answers while ruling out the other two.

This relationship between question structure and experimental design is what scientific elegance actually means. It is not about simplicity for its own sake. A simple experiment that does not cleanly distinguish between competing possibilities is not elegant. It is inadequate. Elegance in experimental science means that the design of the experiment is precisely matched to the logical structure of the question, so that the results cannot be interpreted in multiple ways and the conclusion is forced by the data rather than imposed by the experimenter.

This is a standard that most experiments do not meet and that all experiments should aspire to. When you read about an experiment, in a paper or a textbook, it is worth asking: how many competing hypotheses does this experimental design allow you to distinguish between? If the experiment can only rule out one alternative, it is doing less work than an experiment that rules out two or three alternatives simultaneously. The Meselson-Stahl experiment ruled out two alternatives in two generations of bacteria. That ratio of conclusions to observations is what made it beautiful.

There is a connection between what makes an experiment beautiful and what makes a paper worth reading carefully, and it is worth naming explicitly because it runs through everything this book is trying to do.

A beautiful experiment is one where the design is so precisely matched to the structure of the question that the result cannot be interpreted in more than one way. The elegance is not aesthetic. It is epistemic. It comes from the removal of ambiguity, from the creation of conditions under which nature has only one thing left to say.

Reading a scientific paper carefully requires something similar. You have to hold the question the paper is trying to answer clearly in your mind before you can see whether the paper actually answers it, whether the evidence supports the conclusion, whether the conclusion goes further than the evidence warrants, whether there is something being said quietly that the loud parts of the paper are drawing attention away from. The reader who comes to a paper without the question firmly in mind will extract facts. The reader who comes with the question will see the argument. Meselson and Stahl designed their experiment so that the question structured everything that followed. This book asks you to read their paper, and all the papers in this collection, in the same spirit.

* * *

There is a further lesson in this experiment that is worth drawing out explicitly, because it connects what Meselson and Stahl did to what Watson and Crick had done five years earlier, and to what Hardy had done fifty years before that.

Watson and Crick proposed a structure. The structure implied a mechanism. The mechanism made specific predictions about what should be observable experimentally. Meselson and Stahl tested those predictions. The predictions were confirmed.

This sequence, from structure to mechanism to prediction to experimental test, is one of the fundamental patterns of scientific progress in molecular biology, and understanding it as a pattern, rather than as a collection of disconnected historical episodes, changes how you read the individual papers. Hardy's equation, as we saw in Chapter 1, provided a baseline against which real populations could be measured. The deviation from the baseline was the signal that something interesting was happening.

In the same way, Watson and Crick's structure provided a baseline, a proposed mechanism, against which Meselson and Stahl's experimental results could be measured. The match between prediction and observation was the confirmation that the proposed mechanism was correct.

Science progresses not by accumulating isolated facts but by building structures, whether mathematical structures like Hardy's equation or molecular structures like Watson and Crick's double helix, that generate predictions, and then testing those predictions against observation. The structures that survive the testing are the ones that accumulate into what we eventually call scientific knowledge. The structures that fail the testing are discarded, sometimes quickly, as Pauling's triple helix was discarded within weeks of Watson and Crick's paper, and sometimes slowly and reluctantly, as Delbruck's dispersive replication model was discarded after Meselson and Stahl's centrifuge tubes showed him the data.

* * *

One last thing deserves attention before you read the paper, and it concerns the role of technique in scientific discovery.

Meselson and Stahl's experiment required density gradient centrifugation in caesium chloride solution, a technique that did not exist in a usable form before Meselson developed it. Without that technique, the experiment could not have been done, regardless of how well the question was formulated or how clearly the competing predictions had been worked out. The technique was not just a tool that enabled the experiment. It was, in a real sense, part of the discovery. Developing a new experimental technique is a scientific contribution as significant as developing a new theoretical framework, and it is often less recognised because techniques are less glamorous than theories and because the people who develop them are sometimes not the same people who apply them to the most famous problems.

This is another pattern worth noticing as you read through the history of science. Behind almost every celebrated experimental result, there is a technique that made the result possible, and behind that technique there is usually a person or a group of people who developed it, often without knowing exactly what it would eventually be used to discover. The relationship between technique and discovery in science is not one-directional. Techniques enable discoveries. Discoveries sometimes require new techniques. And new techniques, once developed, often find applications that their developers never imagined, in problems that did not yet exist when the technique was being built.

Meselson developed density gradient centrifugation as a way to answer a specific question about DNA replication. The technique subsequently became one of the most widely used tools in molecular biology, applied to problems ranging from the isolation of plasmids to the separation of RNA species to the analysis of protein complexes. The question that drove its development was important. The technique that the question required turned out to be even more broadly important. This is not unusual. It is the normal way that tools and questions interact in the development of scientific knowledge.

* * *

Now read the paper. It is more technical than the Watson-Crick paper, and some of the details of the centrifugation procedure may require a second reading to follow. Do not let the technical detail distract you from the logical structure of the argument, which is what matters most. The logical structure is this: here are three hypotheses, here are the predictions each hypothesis makes, here is the experiment we designed to test those predictions, and here are the results, which are consistent with one hypothesis and inconsistent with the other two. That structure is present throughout the paper, underneath the technical description of how the centrifuge was operated and how the bands were measured and photographed.

After you have read it, here are the questions worth sitting with.

Meselson and Stahl designed their experiment so that three competing hypotheses would each produce a different and distinguishable result. Think about an experiment you have done or read about in your own biology or chemistry education. How many competing hypotheses did that experimental design allow you to distinguish between? What would have been needed to make it distinguish between more?

The technique of density gradient centrifugation was developed specifically for this experiment but turned out to have applications far beyond it. Can you think of another example, from science or from everyday life, where a tool developed for one specific purpose turned out to be broadly useful for purposes its inventors did not anticipate? What does this suggest about the relationship between solving specific problems and making general contributions?

Delbruck had proposed a version of dispersive replication and was, by some accounts, attached to his model. He was nevertheless immediately convinced by Meselson and Stahl's data. What does it take for a scientist to abandon a hypothesis they have invested in? Is this easy or hard, and what does your answer suggest about the relationship between scientific ideals and human psychology?

* * *

Meselson and Stahl demonstrated that DNA replicates semiconservatively, that each daughter molecule contains one original strand and one newly synthesised strand. They demonstrated this by making old DNA and new DNA physically distinguishable, separating them by density, and reading the result from the position of bands in a centrifuge tube. The experimental logic was so clean that the result was unambiguous from the first generation onward.

But notice what kind of knowledge this is. It is knowledge derived from observation, from measurement, from the physical behaviour of DNA molecules in a density gradient. It required building a technique that did not previously exist, developing conditions under which the question could be asked and answered unambiguously, and reading the result from a physical observable, the position of a band, that was connected to the biological question through a chain of physical and chemical reasoning.

All of this is impressive and important. But there is another way to approach the question of what proteins and other biological molecules can and cannot do, a way that requires no experiment at all in the conventional sense, no centrifuge, no isotope labelling, no physical observable. There is a way of deriving biological knowledge from physical principles alone, from the geometry of chemical bonds and the sizes of atoms, before measuring anything. This approach is older than molecular biology and more fundamental, and it was practised with extraordinary precision by a physicist in Chennai who was not widely known outside his field for many years after he produced his most important result. The next chapter is about that physicist and about what his approach reveals about the relationship between physics and biology that neither field, taken alone, can fully see.

Chapter 5

The Physicist in Chennai Whose Work Became Invisible

Ramachandran, Ramakrishnan and Sasisekharan, 1963

In 1963, a scientist working at the University of Madras produced a piece of work so fundamental to the understanding of protein structure that it is now built into every computational tool used to determine, validate, or predict the three-dimensional shape of a protein molecule anywhere in the world. The tool that won the Nobel Prize in Chemistry in 2024, AlphaFold, uses it. Every protein structure deposited in the Protein Data Bank since the 1970s has been validated against it. Every structural biologist alive today works within the framework it established, so completely that most of them no longer think of it as a contribution from a specific person in a specific place at a specific time. It has become part of the invisible infrastructure of the field, the kind of foundational work that disappears into the tools that depend on it, the way the foundations of a building disappear into the floors and walls constructed above them.

The scientist was Gopalasamudram Narayana Ramachandran, known to colleagues as G. N. Ramachandran, and the piece of work was a paper he published in 1963 with two collaborators, C. Ramakrishnan and V. Sasisekharan, in the Journal of Molecular Biology. The paper introduced what is now called the Ramachandran plot, a two-dimensional map that defines the geometric space available to the backbone of a protein chain. It showed, using nothing more than the known geometry of chemical bonds and the physical principle that two atoms cannot occupy the same space simultaneously, that of all the theoretically possible shapes a protein backbone segment could take, only a small fraction are actually physically possible. The rest are forbidden by geometry.

This chapter is about that map, about the reasoning that produced it, and about what it means that a result this important was produced in postcolonial India in 1963 by a physicist who had moved into structural biology because he found the geometric problems more interesting than anything he had encountered in physics, and who received, for many years, a level of recognition that was considerably less than what the importance of his work deserved.

* * *

To understand what Ramachandran did, you need to understand something about the problem of protein structure that existed in 1963, and specifically about the relationship between the chemistry of proteins and the question of what shapes proteins can take.

A protein is a chain of amino acids linked together by peptide bonds. You know this from your biology and chemistry classes. What the textbook description sometimes does not make sufficiently clear is that the peptide bond itself has a very specific and constrained geometry. Because of the way electrons are distributed in the peptide bond, the bond has partial double-bond character, which means that the six atoms involved in and immediately surrounding the peptide bond are constrained to lie in a single plane. This is called the planarity of the peptide bond, and it is not an approximation or a simplification. It is a direct consequence of the quantum mechanical structure of the bond, and it is essentially absolute under normal conditions.

This planarity constraint has an immediate consequence for protein structure. If the peptide bond and its surrounding atoms must all lie in a plane, then the only freedom of movement available to the protein backbone at each amino acid unit is rotation around the two bonds that flank the peptide bond on either side. These two bonds are called the phi bond, which connects the nitrogen of the amino acid to the alpha carbon, and the psi bond, which connects the alpha carbon to the carbonyl carbon. The angle of rotation around the phi bond is called the phi angle, and the angle of rotation around the psi bond is called the psi angle.

Every amino acid unit in a protein chain therefore has two rotational degrees of freedom, phi and psi, and in principle these angles can each take any value between minus one hundred and eighty degrees and plus one hundred and eighty degrees, giving a vast range of possible backbone conformations. A protein of one hundred amino acids would, if all phi and psi combinations were equally available, have an astronomically large number of possible backbone shapes.

But they are not all equally available. And this is what Ramachandran showed.

* * *

The key insight in Ramachandran's paper was to ask a simple but powerful question: if you specify particular values of phi and psi for an amino acid unit in a protein chain, do the atoms in that unit and the atoms in the neighbouring units come too close to each other?

Atoms have a minimum approach distance, called the van der Waals radius, below which they cannot come without extremely high energy cost. If a particular combination of phi and psi angles would force two atoms in the protein backbone to overlap with each other, that combination is physically impossible under normal conditions. It is not merely unfavourable. It is forbidden.

Ramachandran and his colleagues worked out, systematically and carefully, which combinations of phi and psi angles were geometrically possible and which were forbidden by steric clashes, which is the term used for the situation where two atoms are forced too close together. They did this using the known bond lengths and bond angles of the peptide group and the known van der Waals radii of the relevant atoms. The calculation required no exotic equipment, no particle accelerators, no electron microscopes. It required geometry and the physical principle that two atoms cannot occupy the same space, applied with precision and care to a specific molecular system.

The result was a two-dimensional map, with phi on one axis and psi on the other, showing which regions of the phi-psi space are accessible to protein backbone segments and which are not. The allowed regions turned out to be a small fraction of the total space. Most of the phi-psi plane is forbidden. Protein backbones can only exist in the allowed regions, which correspond to the conformations we recognise as the major secondary structural elements of proteins: the alpha helix, the beta sheet, and a few other less common but structurally important arrangements.

This map is the Ramachandran plot, and it is one of the most information-dense diagrams in all of structural biology. Every point on the plot represents a combination of phi and psi angles. Every point in the allowed regions represents a backbone conformation that is physically possible. Every point in the forbidden regions represents a conformation that cannot exist. And when you plot the phi and psi angles of every amino acid from every known protein structure, they cluster tightly within the allowed regions, confirming that the map correctly identifies the geometric constraints that govern protein backbone geometry.

* * *

It is worth pausing here to understand what kind of reasoning produced the Ramachandran plot, because it is different in important ways from the kinds of reasoning we have examined in the previous four chapters.

Hardy's reasoning was mathematical. He started with a probability model, made explicit simplifying assumptions, and derived algebraic consequences that followed necessarily from those assumptions. The result was an equation that described what must happen in an idealised population under specified conditions.

Watson and Crick's reasoning was structural and inferential. They used experimental data, primarily X-ray diffraction patterns, to constrain the possible structures for DNA, and then argued that one particular structure was consistent with all the available constraints while competing structures were not.

Meselson and Stahl's reasoning was experimental. They designed an experiment to distinguish between competing hypotheses by creating conditions under which each hypothesis made different and testable predictions, and then measured the outcome.

Ramachandran's reasoning was geometric and deductive. He started with known physical facts about bond geometry and atomic radii, applied the principle that atoms cannot overlap, and derived the necessary consequences for the space of possible protein backbone conformations. The result was not a hypothesis to be tested but a constraint that followed necessarily from physics. In principle, Ramachandran could have done this calculation before a single protein structure had been determined experimentally, and the result would have been equally valid.

This is a different kind of biological knowledge from the other three. It is knowledge derived from physical principles rather than from observation or experiment, and it is therefore, in a certain sense, more certain than empirical knowledge. You do not need to measure protein backbone angles to know that most phi-psi combinations are forbidden. You need only understand the geometry of the peptide bond and the physical impossibility of atomic overlap. The Ramachandran plot is a map of biological possibility derived entirely from physics, and its validity does not depend on how many protein structures have been experimentally determined.

This does not make it less biological. It makes it foundational in the specific sense that it defines the space within which all experimental observations of protein structure must fall. Every protein structure determined by X-ray crystallography or cryo-electron microscopy or nuclear magnetic resonance is checked against the Ramachandran plot as a matter of routine. A protein structure in which many amino acid residues fall outside the allowed regions is almost certainly incorrect, because geometry does not permit those conformations. The plot is not a model that might be wrong. It is a constraint that cannot be violated.

* * *

When AlphaFold was trained to predict protein structures, the Ramachandran plot was part of the framework within which the predictions were validated. A predicted structure was evaluated partly on the basis of how well its backbone angles fell within the allowed regions of the Ramachandran plot. This is not a minor technical detail. It means that every protein structure prediction AlphaFold produces is constrained by, and evaluated against, the geometric reasoning that Ramachandran worked out in Chennai in 1963.

The connection between Ramachandran's paper and AlphaFold is therefore not merely historical or contextual. It is structural. Ramachandran's work is literally inside AlphaFold, in the sense that AlphaFold's predictions are constrained by and evaluated against the geometric framework Ramachandran established. Without the Ramachandran plot, there would be no reliable way to assess whether a predicted protein structure is geometrically plausible, which means there would be no reliable way to train or validate AlphaFold's predictions.

This is what it means for a piece of scientific work to become invisible infrastructure. Ramachandran's plot is not cited in AlphaFold's paper the way a recent result would be cited, with a discussion of its specific contribution. It is part of the background knowledge of structural biology, assumed rather than stated, present in the validation framework rather than in the main argument. The work has become so foundational that it is no longer experienced as a contribution from a specific person. It is simply the way things are done.

* * *

The question of why Ramachandran's work received less recognition than its importance deserved is one that does not have a single simple answer, but it has several partial answers that are worth examining, because they illuminate the same mechanisms of recognition and obscurity that we saw in the Hardy-Weinberg story in Chapter 2.

Ramachandran was working in India in 1963. India had been independent for only sixteen years. Its scientific institutions were still being built, and they were being built in conditions of limited resources and limited international connectivity. The major centres of structural biology were in Britain and the United States, where X-ray crystallography was being developed into the powerful technique it would become, and where the community of scientists working on protein structure was concentrated. Ramachandran was outside that network, geographically and institutionally.

He published in English, which meant the language barrier did not prevent his work from being read. But the journal he published in, while legitimate and international, did not have the same reach and prestige as Nature or the journals published by American scientific societies. The people who needed to know about his work did know about it, eventually, but the speed and completeness with which a result becomes embedded in a field's practice depends significantly on who produces it and where, and Ramachandran's location worked against him in ways that had nothing to do with the quality of the science.

There is also a disciplinary factor. Ramachandran was a physicist who had moved into structural biology. The structural biology community of the 1960s was dominated by chemists and crystallographers, and a physicist's contribution, framed in the language of geometry and physical constraints rather than in the language of crystallographic analysis, may have seemed to some members of that community to be less central to their enterprise than it actually was. Disciplinary boundaries shape the reception of ideas in ways that are difficult to disentangle from the ideas themselves.

What is clear, looking back from the present, is that the Ramachandran plot is one of the most widely used analytical tools in structural biology, that it was produced by a scientist working in postcolonial India with limited resources and limited international connectivity, and that the recognition it received was for many years less than proportional to its importance. This is not a story with a villain. It is a story about the systematic operation of factors, geography, institutional prestige, disciplinary boundaries, that shape scientific recognition in ways that are independent of scientific merit.

For a student reading this book anywhere in the world outside the major centres of global science, this story is not merely historically interesting. The factors that shaped Ramachandran's recognition in 1963 are still operating today, in modified but recognisable forms. Understanding them is part of understanding how science actually works, as opposed to how it would work in an ideal world where merit and recognition always corresponded perfectly.

I read Ramachandran's story differently from how I might have read it if I were writing this book from a different place.

From Kuchinda, the outline of his situation is not unfamiliar. The work that is foundational but not immediately visible as foundational to the people who determine what counts as foundational. The institution whose name does not open the doors that names open. The contribution that gets used before it gets credited, that becomes part of the infrastructure of a field before it becomes part of the field's official story of itself. These are not abstractions. They are recognisable conditions, experienced in different degrees by scientists working from institutions that the centre of global science does not automatically attend to.

I am not claiming equivalence between Ramachandran's situation and my own, or between his contribution and anything I have done. The comparison is not about scale. It is about structure. The structure of his position, the physicist in Chennai in 1963 whose geometric reasoning became invisible infrastructure for a field that did not immediately know where to place him, has a shape that is legible from where I am sitting in a way that it might not be legible from Cambridge or London or Caltech.

This is why Ramachandran is not incidental to this book. He is not here because including a non-Western scientist makes the collection more representative, though it does. He is here because his paper is genuinely foundational, because his story is genuinely instructive, and because reading his story from Kuchinda rather than from one of the institutions where the other papers were written changes what the story is about. From here it is not a story about an exceptional scientist who overcame geographical disadvantage. It is a story about what science looks like when you are inside the condition rather than observing it from outside.

* * *

There is one more dimension to Ramachandran's story that deserves attention, and it concerns the relationship between his training as a physicist and the nature of his contribution to biology.

Ramachandran came to structural biology from physics, specifically from crystallography and optics, and he brought with him a physicist's instinct for working from first principles, for asking what the fundamental constraints on a system are before asking what the system actually does. This instinct is what produced the Ramachandran plot. A biologist approaching the protein structure problem would naturally have started by measuring protein structures and then looking for patterns in the measurements. A physicist approaching the same problem would naturally have started by asking what physical principles constrain the possible structures, before looking at any measurements at all.

Both approaches are valid, and in this case both approaches were being pursued simultaneously by different groups. But the physicist's approach produced a more fundamental result, precisely because it derived constraints from first principles rather than extracting patterns from data. A constraint derived from first principles is not provisional in the way that an empirical pattern is provisional. It does not become less valid when more data are collected. It becomes more confirmed as more protein structures are determined and more backbone angles are measured and found to fall within the predicted allowed regions.

This is another instance of the pattern we saw with Hardy in Chapter 1. Hardy was not a biologist. He was a mathematician who brought a mathematician's instinct for clarity and generality to a biological problem that biologists were struggling to see clearly. Ramachandran was not a structural biologist in the conventional sense. He was a physicist who brought a physicist's instinct for first principles to a problem that structural biologists were approaching empirically. In both cases, the outsider's perspective produced a more fundamental result than the insiders were producing, not because the outsider was smarter but because the outsider was asking a different kind of question.

This is a pattern worth watching for, in the history of science and in your own intellectual work. The person who does not know that a problem is supposed to be approached in a certain way is sometimes the person who approaches it in the right way. The disciplinary training that makes you expert in a field also gives you assumptions about how problems in that field should be attacked, and those assumptions are sometimes the obstacle to seeing the most direct solution. Ramachandran did not know, in the way that an established structural biologist would have known, that the standard approach to protein structure was to measure structures and look for patterns. He asked instead what physics permitted, and the answer was the Ramachandran plot.

* * *

Now read the paper. It is more technical than any of the papers you have read so far in this book, and some of the notation and terminology will require patient reading. The key things to follow are the argument about the planarity of the peptide bond and why it matters, the derivation of the allowed and disallowed regions from van der Waals contacts, and the comparison of the predicted allowed regions with the backbone angles observed in actual protein structures known at the time. The figures in the paper are the Ramachandran plot itself, and they deserve careful attention. The allowed regions in the plot correspond to the alpha helix and beta sheet conformations you know from your biology classes. Seeing those familiar structural elements emerge from a geometric argument about atomic radii is one of the more satisfying moments this paper has to offer.

After you have read it, here are the questions worth sitting with.

Ramachandran derived the Ramachandran plot from first principles, before most protein structures had been experimentally determined. In principle, he could have published the plot in 1950, or 1940, if the geometry of the peptide bond and the van der Waals radii of the relevant atoms had been known. What does it mean that a result that could have been derived earlier was not derived earlier? What was needed, beyond the technical knowledge, for the question to be asked?

The Ramachandran plot defines what protein structures cannot be. It is a map of impossibility rather than a map of what actually exists. Why is a map of impossibility scientifically useful? Can you think of other examples in science or mathematics where defining what is impossible has been more productive than cataloguing what is possible?

Ramachandran's work became invisible infrastructure, present in every structural biology tool but rarely cited explicitly as a recent contribution. Is this a problem? Is there a difference between a contribution being used and a contribution being recognised, and does the difference matter for how science as an institution functions?

* * *

Ramachandran derived the constraints on protein backbone geometry from first principles, before most protein structures had been experimentally determined, using nothing more than the known geometry of chemical bonds and the physical impossibility of atomic overlap. The result was a map of what protein backbones can and cannot be, a map whose validity does not depend on how many protein structures have been measured because it follows necessarily from physics rather than from observation.

This is knowledge of a particular and powerful kind: knowledge that constrains possibility before experience, that tells you what the world cannot contain before you look. It is the kind of knowledge that physicists have been producing for centuries, from the conservation laws to the Pauli exclusion principle, and it is rare in biology because biology is primarily an empirical science, a science that reads the world by measuring it rather than by deriving what it must be from prior principles.

But something changed in biology in the last decade of the twentieth century and the first two decades of the twenty-first. The amount of biological data being produced, protein structures, genome sequences, gene expression patterns, protein interaction networks, grew faster than the ability of human minds to extract meaning from it by conventional empirical analysis. And into this situation came a different kind of tool, one that did not derive knowledge from first principles or extract it from carefully designed experiments but learned it from examples, from the accumulated record of what biology has actually produced over billions of years of evolution. The next chapter is about that tool and about the genuinely new kind of knowledge it produces, knowledge that is real and accurate and powerful but that no human mind can fully explain or fully understand.

Chapter 6

What the Machine Learned That No One Can Fully Explain

Jumper et al., AlphaFold, 2021

In the autumn of 2020, a team of researchers from DeepMind, an artificial intelligence laboratory owned by Alphabet, the parent company of Google, entered a competition called the Critical Assessment of protein Structure Prediction, known as CASP. This competition has been held every two years since 1994, and its purpose is to assess the accuracy of computational methods for predicting the three-dimensional structure of a protein from its amino acid sequence alone, without using any experimental data about that particular protein's structure. The competition works by giving participants the sequences of proteins whose structures have been recently determined experimentally but not yet made public, asking participants to predict those structures, and then comparing the predictions against the actual experimental structures when they are released.

CASP had been running for twenty-six years by the time DeepMind entered in 2020, and over those twenty-six years the accuracy of the best prediction methods had improved steadily but slowly. The gap between computational predictions and experimental structures, measured on a standard accuracy scale running from zero to one hundred, had narrowed from the early days of the competition but remained significant. The best methods were achieving scores in the range of forty to sixty on the most difficult prediction targets, which structural biologists considered reasonable progress but far short of what would be needed to make computational structure prediction genuinely useful as a substitute for experimental determination.

DeepMind's entry, which they called AlphaFold2 to distinguish it from an earlier version they had entered in the 2018 competition, achieved a median score of ninety-two point four on the most difficult targets. The next best entry achieved a median score of around fifty. The gap between AlphaFold2 and the second-best method was larger than the gap between the second-best method and the worst method in the competition. Several of the experimentally determined structures that served as the benchmark were matched by AlphaFold2 predictions with an accuracy that was, in several cases, within the margin of experimental error of the structure determination itself. In other words, AlphaFold2 was predicting protein structures about as accurately as the experimental techniques used to determine them.

The structural biology community's reaction to these results ranged from astonishment to something closer to vertigo. The protein folding problem, the challenge of predicting a protein's three-dimensional structure from its sequence, had been an open problem in biology for more than fifty years. It had resisted every approach that had worked for other hard problems in molecular biology. It had defeated theoretical physicists, computational chemists, and bioinformaticians. And then a team of machine learning engineers, most of whom did not have biology as their primary training, had solved it, or something very close to solving it, using an approach that was in several important respects unlike anything that had been tried before.

This chapter is about what AlphaFold actually does, why it works when fifty years of other approaches did not, and what kind of knowledge it produces. That last question is the most important one, and the most difficult to answer, because AlphaFold produces knowledge in a way that is genuinely new in the history of science, different in kind from anything produced by Hardy's algebra, Watson and Crick's model building, Meselson and Stahl's centrifuge tubes, or Ramachandran's geometric reasoning. Understanding what is new about it, and what that newness means for how we think about scientific knowledge, is what this chapter is for.

* * *

To understand why AlphaFold succeeded where fifty years of other approaches failed, you need to understand what those other approaches were trying to do and why the problem resisted them.

The protein folding problem, in its original formulation, was a physical problem. A protein is a chain of amino acids, and when it is synthesised in a cell it folds into a specific three-dimensional shape determined by its amino acid sequence. The folded shape is the shape in which the protein is thermodynamically most stable, meaning it is the shape in which the total energy of the system, including all the interactions between amino acids and between the protein and the surrounding water molecules, is minimised. If you could calculate the energy of every possible conformation of a protein and find the one with the lowest energy, you would have solved the protein folding problem.

The difficulty is that the number of possible conformations of even a small protein is astronomical. A protein of one hundred amino acids has, if you allow for the rotational freedom at each amino acid's backbone bonds and side chain bonds, something like ten to the power of three hundred possible conformations. Searching this space exhaustively is computationally impossible. The universe is not old enough, and no computer is fast enough. This is called Levinthal's paradox, named after the biophysicist Cyrus Levinthal who pointed it out in 1969: if a protein found its folded structure by randomly sampling conformations, it would take longer than the age of the universe. Yet proteins fold in milliseconds. They must therefore be following some kind of guided pathway rather than searching randomly, but for fifty years nobody could work out what that pathway was well enough to simulate it computationally with sufficient accuracy to predict the final structure reliably.

Various approaches were tried. Some groups tried to simulate the physical folding process directly, using molecular dynamics calculations that tracked the movement of every atom in the protein and the surrounding water over time. This was computationally expensive and, for large proteins, still too slow to be practical even with modern computers. Other groups tried to use evolutionary information, reasoning that amino acids in a protein that are spatially close to each other in the folded structure tend to evolve together, because a mutation in one that disrupts their interaction is likely to be compensated by a mutation in the other. By analysing the patterns of correlated mutations in large collections of related protein sequences, it was possible to infer which pairs of amino acids were likely to be spatially close, and this information could be used to constrain the structure prediction. This approach, called coevolutionary analysis, improved prediction accuracy significantly but still fell far short of what AlphaFold eventually achieved.

What AlphaFold did was different from both of these approaches. It did not try to simulate the physical folding process. It did not try to derive the structure from first principles, the way Ramachandran had derived the Ramachandran plot. It learned, from a database of approximately one hundred and seventy thousand protein structures that had been experimentally determined over the preceding fifty years and deposited in the Protein Data Bank, what protein structures look like. More precisely, it learned the relationship between protein sequences and protein structures, and it learned this relationship at a level of detail and generality that no human mind could have extracted from the same data by conscious reasoning.

* * *

The system that does this learning is called a neural network, and understanding what a neural network is and what it does is important for understanding both the power and the limits of AlphaFold.

A neural network is a computational system loosely inspired by the structure of the brain, consisting of many layers of simple computational units connected to each other, in which the strength of the connections between units, called the weights, is adjusted during a training process so that the network produces correct outputs when given particular inputs. In AlphaFold's case, the inputs are protein sequences and related evolutionary information, and the outputs are predicted three-dimensional coordinates for every atom in the protein. The training process involves showing the network many examples of sequences whose structures are known, adjusting the weights so that the network's predictions match the known structures, and repeating this process millions of times across hundreds of thousands of examples until the network's predictions are sufficiently accurate.

The result of this training process is a network with hundreds of millions of parameters, each of which has been tuned to a specific value by the training procedure. These parameters collectively encode, in a distributed and implicit way, the relationship between protein sequences and protein structures as it is manifested across the entire Protein Data Bank. But they encode this relationship in a form that no human can read or interpret directly. You cannot look at the weights of a trained neural network and explain, in biological terms, why the network makes the predictions it makes. The knowledge is there, demonstrably, because the predictions are accurate. But it is not there in any form that is accessible to conscious human understanding.

This is what is genuinely new about AlphaFold as a form of scientific knowledge. Hardy's equation can be understood completely. You can follow every step of the derivation and explain exactly why the result is what it is. Watson and Crick's double helix can be understood completely. You can explain why the base pairing rules are what they are, why the strands are antiparallel, why the helix has the dimensions it has. Meselson and Stahl's result can be understood completely. You can follow the experimental logic and explain exactly why the band patterns rule out two of the three models. Ramachandran's plot can be understood completely. You can explain, from the geometry of the peptide bond and the van der Waals radii of the atoms, exactly why each region of the phi-psi plane is allowed or forbidden.

AlphaFold's predictions cannot be understood in this way. You can describe the architecture of the network, the training procedure, and the general nature of what the network has learned. But you cannot, for any specific prediction, explain in biological terms why the network predicted that particular structure. The prediction is accurate. The reason for the accuracy is not fully accessible to human understanding.

* * *

The specific architectural innovations that make AlphaFold work are worth describing, even though the details are technically demanding, because they reveal something important about how the problem was eventually cracked.

The central component of AlphaFold is a module called the Evoformer, a name that reflects the two kinds of information it processes: evolutionary information, in the form of multiple sequence alignments showing how the protein sequence varies across many related species, and pairwise information about the relationships between pairs of amino acid positions in the sequence. The Evoformer processes these two kinds of information simultaneously, allowing patterns in the evolutionary variation to inform the pairwise relationship predictions and vice versa, across forty-eight repeated layers of computation.

The evolutionary information is particularly important. When AlphaFold looks at a protein sequence, it does not see just that sequence. It sees the sequence in the context of hundreds or thousands of related sequences from different species, showing which positions in the sequence have been conserved across evolution and which have varied. Positions that have been conserved are likely to be structurally or functionally important. Positions that tend to vary together are likely to be spatially close in the folded structure, because their coevolution reflects a structural constraint. The Evoformer learns to extract this information from the multiple sequence alignment and use it to constrain the structure prediction.

The output of the Evoformer feeds into a structure module that translates the learned representations into actual three-dimensional coordinates. The structure module introduces an explicit geometric framework, representing each amino acid as a rigid body that can rotate and translate in three-dimensional space, and it uses the information from the Evoformer to determine the position and orientation of each rigid body. The Ramachandran plot appears here, as part of the framework within which the backbone geometry is constrained during training and evaluation.

The entire system is trained end-to-end, meaning that the parameters of both the Evoformer and the structure module are adjusted simultaneously during training, so that the two components learn to work together optimally. This end-to-end training is itself an important innovation. Earlier computational approaches to protein structure prediction used separate components for different parts of the problem, each trained or optimised independently. AlphaFold treats the entire prediction problem as a single learning task, which allows the system to discover relationships between the different parts of the problem that would not be visible if the parts were handled separately.

* * *

There is a sentence in AlphaFold's paper, published in Nature in 2021, that deserves the same careful attention we gave to Watson and Crick's sentence about the copying mechanism of DNA.

The sentence is this: we hope that AlphaFold and computational approaches that apply its techniques for other biophysical problems will become essential tools of modern biology.

Compare this to Watson and Crick's sentence: it has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material.

The two sentences are doing similar things in similar ways. Both are pointing, quietly and with deliberate restraint, toward implications that go far beyond the immediate result. Watson and Crick are pointing toward the entire mechanism of heredity. AlphaFold's authors are pointing toward the possibility of solving a vast range of biological problems, from drug design to the understanding of disease mechanisms to the engineering of new proteins with desired properties, using approaches like the one they have demonstrated.

But there is a difference between the two sentences that reveals something important about how science has changed in the sixty-eight years between them. Watson and Crick's sentence is in the present tense, pointing to something that their result immediately suggests. It is a logical implication of the structure they have proposed. AlphaFold's sentence is in the future tense, expressing a hope rather than identifying an implication. It is more cautious, more tentative, more aware of the distance between a demonstrated result and the applications that result might eventually enable.

This difference in tone reflects a genuine difference in what the two papers have actually shown. Watson and Crick's structure logically implies a copying mechanism. The implication is direct and necessary, which is why the sentence can be written in the present tense. AlphaFold's success at predicting protein structures does not logically imply that the same approach will work for other biophysical problems. The hope is reasonable, and subsequent developments have confirmed that it is more than just a hope, but at the time of writing it was still a hope rather than a demonstrated fact. The authors chose their tense carefully, as Watson and Crick chose their word suggests rather than proves.

* * *

It is also worth being honest about what AlphaFold does not do, because the enthusiasm that greeted its results in 2021 sometimes obscured the distinction between what was demonstrated and what remained to be solved.

AlphaFold predicts the structure of individual protein chains, and it does this with remarkable accuracy for proteins that have many related sequences in the database. For proteins with few related sequences, its accuracy decreases significantly. For protein complexes, meaning proteins that function as assemblies of multiple chains, the original AlphaFold was less reliable, though subsequent versions have addressed this limitation. For intrinsically disordered proteins, meaning proteins that do not adopt a single stable three-dimensional structure but rather exist as ensembles of interconverting conformations, AlphaFold's single-structure prediction framework is not the right tool.

More fundamentally, AlphaFold predicts structure but does not explain why the structure is what it is. It cannot tell you which amino acid interactions are responsible for the stability of the predicted structure, which parts of the structure are rigid and which are flexible, or how the structure changes when the protein binds to another molecule or undergoes a chemical modification. These are questions that require physical understanding, not pattern recognition, and they remain as difficult as they ever were. AlphaFold has solved the structure prediction problem, or most of it. It has not solved the protein physics problem, which is deeper and harder and requires a different kind of knowledge from the kind AlphaFold provides.

This distinction between prediction and explanation is important not just for AlphaFold but for the broader question of what artificial intelligence can and cannot contribute to science. A system that can predict accurately without explaining is genuinely useful. Many important things in biology can be learned from knowing structures, even without knowing why those structures are what they are. But a system that can only predict, and cannot explain, cannot by itself generate the kind of mechanistic understanding that allows scientists to design new experiments, build new theories, or understand biological processes at the level of physical causation. Prediction and explanation are complementary forms of knowledge, and science needs both.

There is something worth sitting with before moving to the synthesis chapter, something that AlphaFold makes visible that the other papers in this collection do not make visible in quite the same way.

Every other paper in this book was produced by a human mind that could, at least in principle, explain every step of its own reasoning. Hardy could walk you through the algebra. Watson and Crick could show you why the base pairing geometry works. Meselson could explain why a band at an intermediate density position rules out conservative replication. Ramachandran could derive the allowed regions of the phi-psi plane from first principles, starting from the geometry of the peptide bond and the van der Waals radii of the relevant atoms, in front of you, on a blackboard, without notes.

AlphaFold cannot do this. Not because the people who built it are less intelligent than Hardy or Ramachandran. Because the knowledge is not in any human mind. It is distributed across hundreds of millions of parameters, each tuned by a training process that processed more structural information than any human being could consciously examine in a lifetime. The knowledge is real. The predictions are accurate. But the reasoning that produces them is not available to human inspection in the way that Hardy's algebra or Ramachandran's geometry is available.

I find this genuinely strange, and I think it is worth saying so directly rather than absorbing it into a smooth analytical account. We have built a system that knows something we do not know and cannot fully explain. It reads biological information at a level of depth and scale that no human mind can reach alone. And it does so, as Chapter 7 will argue, by reading the evolutionary history written into protein sequences, which means it reads evolution in order to read structure, which means, in a sense that is not metaphorical, that evolution is reading itself.

That is where the six papers have brought us. From Hardy's algebra in 1908 to a machine reading evolution in 2021. The distance between those two points is the distance this book has been travelling.

* * *

Now bring together everything this book has shown you, because this is the moment to see how the six papers connect.

Hardy showed that the allele frequencies in a population carry information about evolutionary forces, and that a simple mathematical model can extract that information by defining a baseline against which deviations are measured. The information was always in the allele frequencies. Hardy provided the framework for reading it.

Watson and Crick showed that the sequence of bases along a DNA strand carries information sufficient to specify a complementary strand, and that this complementarity is the mechanism of hereditary transmission. The information was always in the sequence. Watson and Crick showed what it was information about and how it was read.

Meselson and Stahl showed that the mechanism Watson and Crick proposed was correct, by designing an experiment whose results were interpretable in only one way. The mechanism was always operating in every dividing cell. Meselson and Stahl found a way to make it visible.

Ramachandran showed that the geometry of the peptide bond encodes a constraint on the space of possible protein structures, and that this constraint is so fundamental that it governs every protein in every organism. The constraint was always there, written in the physics of the bond. Ramachandran read it.

AlphaFold showed that the sequence of amino acids in a protein carries information sufficient to specify, with near-experimental accuracy, the three-dimensional structure of that protein, and that this information can be extracted by a system that has learned, from hundreds of thousands of examples, the relationship between sequence and structure. The information was always in the sequence. AlphaFold found a way to read it that no human could have found by conscious reasoning alone.

The argument that runs through all five of these papers, and through Edwards' historical analysis of how the Hardy-Weinberg principle came to be, is the same argument stated in five different experimental languages. Biology is information. The information has structure. The structure can be read, by algebra, by model building, by experiment, by geometry, by machine learning. And each new way of reading it reveals something that the previous ways could not reach.

* * *

After you have read AlphaFold's paper, or the sections of it reproduced in this chapter, here are the questions worth sitting with.

AlphaFold produces accurate predictions whose basis it cannot explain. Hardy's equation produces accurate predictions whose basis can be fully explained. Is one of these kinds of knowledge better than the other? What would you need to know about a specific biological problem in order to decide which kind of knowledge would be more useful for addressing it?

AlphaFold was built by engineers and machine learning researchers, most of whom were not structural biologists by training. The protein folding problem had resisted structural biologists for fifty years. Is there a pattern here that connects to what we saw with Hardy and Ramachandran, both of whom were outsiders to the fields whose problems they solved? What does this pattern suggest about the value of disciplinary boundaries in science?

The Ramachandran plot, derived in 1963 from first principles by a physicist in Chennai, is present inside AlphaFold as part of its validation framework. What does this connection across fifty-eight years tell you about the timescale on which foundational scientific contributions become visible in their full importance? And what does it suggest about how you should think about the scientific work being done right now, in places and by people that the major centres of science may not yet be paying full attention to?

* * *

Six papers. Five primary papers and one historical analysis. Mathematical reasoning, social and historical reasoning, structural reasoning, experimental reasoning, geometric reasoning, computational reasoning. Six different ways of reading biological information at six different levels of organisation, from the population to the genome to the molecule to the protein backbone to the sequence itself.

Each paper reads something different. But they are all reading the same world, and the world they are reading has a property that none of the papers individually explains but that all of them together make visible. The biological world is a world in which information exists at every level of organisation, in which that information is structured into readable patterns, and in which the methods that read it are adapted, with extraordinary precision, to the specific structure of the information they extract. This is not a coincidence. It is not an artefact of how we have chosen to study biology. It is a feature of the biological world itself, and it has an explanation.

The next chapter is about that explanation. It is the chapter in which all six papers are read together, as stages in a single argument that none of them could have made alone, and in which the argument arrives at a conclusion that was implicit in every chapter but could not be stated until all six had been read. It is also the chapter in which Dobzhansky's most famous sentence, the sentence that every biology student encounters and almost nobody thinks about carefully, is extended in a direction that the six papers make possible and that biology, taken alone, cannot reach.

Chapter 7

One Conversation Across One Hundred and Fifty-Eight Years

Synthesis

The argument this chapter makes was not designed in advance. It emerged from reading the six papers carefully and asking, after each one, what it had in common with the others that none of them stated explicitly. The answer took a long time to become visible. It is the kind of answer that cannot be seen until you have read everything that leads to it, which is why it appears here, in the seventh chapter, rather than in the preface or the introduction.

This matter because the argument is genuinely the product of the reading rather than a thesis the reading was assembled to support. There is a difference between those two things, and the difference is not merely rhetorical. A thesis assembled in advance selects evidence that confirms it and arranges that evidence efficiently. An argument that emerges from reading arrives with the texture of the reading still on it, with the roughness of ideas that were discovered rather than demonstrated, with the particular shape that comes from following a question wherever it leads rather than leading it where you want it to go.

What follows is the latter kind of argument. It arrived at the end of a long reading, not at the beginning of a planned demonstration. Whether it is correct is for the reader to judge. But it is offered in the spirit in which it was found, which is the spirit of someone who was trying to understand something and found, at the end of trying, that the understanding had a shape they had not anticipated when they started.

* * *

There is a quotation that every biology student encounters at some point in their education, usually in the context of a lecture on evolution or in the introductory chapter of a genetics textbook. It was written by the Ukrainian-American geneticist Theodosius Dobzhansky in 1973, and it reads: nothing in biology makes sense except in the light of evolution. The quotation has become so widely repeated that it has acquired the slightly worn quality of a phrase that everyone knows and nobody thinks about carefully anymore. It is cited as a truism, as a way of gesturing toward the importance of evolutionary thinking without actually doing any evolutionary thinking, and in this overuse something important about what Dobzhansky actually meant has been lost.

What Dobzhansky meant was precise and radical. He was not saying that evolution is an important topic in biology, one subject among many that deserves attention. He was saying that evolution is the explanatory framework without which no biological fact makes sense as a fact rather than as an isolated observation. The genetic code is universal across all living organisms. That is a fact. But it is an incomprehensible fact, a fact that just sits there demanding explanation, unless you understand that all living organisms share a common ancestor and that the genetic code was established early in the history of life and then conserved because any change to it would be catastrophic for an organism whose entire biochemistry depended on it. The universality of the genetic code makes sense only in the light of evolution. Without evolution, it is merely strange.

Six papers. One hundred and thirteen years. And by the time you have read all of them, you are in a position to see something that Dobzhansky's quotation points toward but does not quite reach. You are in a position to see that his insight can be extended in a direction that nobody, as far as I am aware, has stated clearly before.

The extension is this: not only does nothing in biology make sense except in the light of evolution, nothing in the history of how we read biology makes sense except in the light of an analogous process that operates on scientific methods and frameworks the way evolution operates on genes and organisms.

That is the argument of this chapter. It will take some time to develop properly, and it requires holding several ideas in mind simultaneously, but every piece of it is built from things you have already encountered in the six chapters that came before. This chapter does not introduce new evidence. It shows you what the evidence you already have, taken together, actually means.

* * *

Begin with the most basic question the six papers raise collectively. Why is biological information readable at all?

This question sounds strange at first because we are so accustomed to the fact of readability that we have stopped noticing it needs explanation. Hardy could read evolutionary information from allele frequencies using algebra. Watson and Crick could read hereditary information from the geometry of the double helix. Meselson and Stahl could read the replication mechanism from the density distribution of DNA molecules in a centrifuge tube. Ramachandran could read the constraints on protein structure from the geometry of the peptide bond. AlphaFold could read the three-dimensional structure of a protein from its amino acid sequence and the evolutionary history of related sequences. Each of these readings was possible because biological information, at the level being examined, had a structure that the method being applied could extract.

But biological information did not have to be structured this way. There is no logical necessity that allele frequencies should carry a signal about evolutionary forces that algebra can extract. There is no logical necessity that the geometry of a molecule should encode the mechanism of its own reproduction. There is no logical necessity that the sequence of amino acids in a protein should contain enough information to specify its three-dimensional structure. These are facts about the biological world, but they are not logical necessities. They could, in principle, have been otherwise.

Why are they the way they are? The answer, and this is the first step in the argument, is evolution.

Allele frequencies carry a readable signal about evolutionary forces because populations whose allele frequencies did not respond to selection, drift, mutation, and migration in a structured way would not have been populations in which evolution could operate effectively, and such populations would not have persisted. The readability of population-level information is a product of the fact that populations which could be shaped by evolutionary forces were the ones that survived and diversified.

The geometry of the double helix encodes the mechanism of its own reproduction because molecules that could not reliably copy themselves were not inherited, and therefore did not accumulate over evolutionary time. The self-specifying complementarity of DNA base pairing is not a coincidence. It is the property that made DNA the hereditary molecule of all life on Earth, precisely because a molecule with this property could be reliably copied and reliably inherited. The readability of molecular structural information is a product of selection for molecules that could encode and transmit information reliably.

The sequence of amino acids in a protein contains enough information to specify its three-dimensional structure because proteins whose sequences did not reliably specify their structures would not fold consistently, would not perform their functions consistently, and would not have been conserved by selection. The sequence-to-structure relationship that AlphaFold reads is not a property of protein chemistry in the abstract. It is a property of the specific protein sequences that evolution has preserved because they fold reliably into functional structures. The readability of sequence information is a product of selection for sequences that fold reliably.

In each case, the readability of the information is not a given. It is a product. It is the product of billions of years of selection operating on variation, preserving the variants whose information was structured in ways that the biological machinery, the ribosomes, the polymerases, the chaperones, the regulatory networks, could read and act on. The biological world is readable because life itself is a reading process, and the information in living systems has been shaped by selection into forms that the reading machinery of life can extract and use.

This is Dobzhansky's insight stated at a deeper level. Not just that the facts of biology make sense in the light of evolution. The readability of biology, the very property that makes scientific investigation of biological systems possible, is itself a product of evolution.

* * *

Now take the second step, which is the more original and more unsettling one.

Each of the five primary papers in this book, Hardy's algebra, Watson and Crick's model building, Meselson and Stahl's experimental design, Ramachandran's geometric reasoning, AlphaFold's machine learning, represents a method of reading biological information. These are not the only possible methods. They are the methods that were actually used, by the specific scientists involved, working in the specific historical and intellectual contexts they inhabited. But why these methods? Why algebra rather than geometry for Hardy's problem? Why model building rather than pure computation for Watson and Crick? Why density gradient centrifugation rather than some other physical technique for Meselson and Stahl?

The answer in each case involves a combination of the specific demands of the problem, the tools available at the time, and the intellectual traditions within which the scientists had been trained. Hardy used algebra because he was a mathematician and because the population genetics problem, once correctly framed, was an algebraic problem. Watson and Crick used model building because Pauling had demonstrated that model building was a powerful approach to molecular structure problems, and because the X-ray data provided constraints that a model had to satisfy rather than a complete specification of the structure. Meselson and Stahl used density gradient centrifugation because Meselson had developed the technique and because the density difference between heavy and light DNA provided a physical observable that could distinguish between the competing hypotheses. Ramachandran used geometric reasoning because he was a physicist trained in crystallography and optics, fields in which deriving constraints from first principles is a standard and powerful approach. DeepMind used machine learning because machine learning had, by 2016, demonstrated the ability to extract complex patterns from large datasets in ways that traditional computational approaches could not match, and because the Protein Data Bank provided a dataset large enough to train a powerful model.

In each case, the method was not chosen from a complete menu of all possible methods. It was chosen from the methods available within the intellectual tradition of the person or group doing the work. And those intellectual traditions did not arise arbitrarily. They are the products of a long history of methods being tried, found useful, refined, extended, and passed on, and methods being tried, found inadequate, and abandoned or modified. The algebra that Hardy used had been developed and refined over centuries because it reliably produced useful results in many domains. The model-building approach that Watson and Crick used had been developed and refined by Pauling and others because it worked for molecular structure problems. The experimental philosophy that Meselson and Stahl embodied, the idea of designing experiments that cleanly distinguish between competing hypotheses, had been developed and refined since at least the seventeenth century because it reliably produced unambiguous results. The geometric first-principles approach that Ramachandran used was a physicist's habit of mind that had been selected for in the intellectual tradition of physics because it produced foundational results. The machine learning approach that DeepMind used was the product of decades of development in artificial intelligence and statistics, preserved and extended because it worked in domain after domain.

The methods that appear in these six papers are not inventions. They are inheritances. Each scientist inherited a set of approaches, a set of tools, a set of ways of framing problems and evaluating solutions, from the intellectual tradition they had been trained in. And those intellectual traditions are themselves the products of something that operates, at a cultural and historical level, the way natural selection operates at a biological level. Methods that produce reliable knowledge are preserved and transmitted. Methods that do not produce reliable knowledge are modified or abandoned. The intellectual traditions that scientists inherit are the accumulated residue of this selection process operating over centuries on the space of possible approaches to understanding the natural world.

This is not a metaphor. Or rather, it is more than a metaphor. It is a structural parallel that deserves to be taken seriously as a claim about how scientific knowledge develops.

Biological evolution proceeds by variation and selection. Random mutations generate variation in the population of organisms. Selection filters that variation, preserving variants that are more fit and eliminating variants that are less fit. The result, over many generations, is a population that is better adapted to its environment than the ancestral population, carrying more information about the structure of the environment in the structure of its genomes and phenotypes.

The evolution of scientific methods proceeds by an analogous process. Variations on existing methods are tried, in new domains or with new modifications. Some of these variations produce reliable knowledge, and they are preserved, taught to students, written into textbooks, extended by subsequent researchers. Others fail to produce reliable knowledge, and they are modified or abandoned. The result, over many generations of scientists, is a repertoire of methods that is better adapted to the task of reading the natural world, carrying more information about the structure of effective inquiry in the structure of its practices and frameworks.

Hardy's algebra is adapted to the task of extracting population-level evolutionary information because it is the product of centuries of selection for algebraic methods that reliably solve probability and combinatorics problems. Watson and Crick's model building is adapted to the task of extracting molecular structural information because it is the product of decades of selection for methods that reliably use geometric and chemical constraints to determine molecular structure. Meselson and Stahl's experimental design is adapted to the task of distinguishing between competing mechanistic hypotheses because it is the product of centuries of selection for experimental approaches that produce unambiguous results. Ramachandran's geometric reasoning is adapted to the task of deriving structural constraints from physical principles because it is the product of centuries of selection for first-principles reasoning in physics. AlphaFold's machine learning is adapted to the task of extracting complex patterns from large datasets because it is the product of decades of selection for learning algorithms that scale effectively with data.

The methods are adapted to the problems. The adaptation is not designed. It is the result of a selection process operating on the population of methods available to scientists at any given time.

* * *

Now bring these two steps together, because together they generate something that neither step generates alone.

Biological information is readable because evolution has structured it into readable forms. Scientific methods are adapted to reading biological information because an analogous selection process has shaped them into forms that work. This means that the readability of biological information and the adaptedness of the methods used to read it are both products of evolutionary processes, one biological and one cultural, that have been operating in parallel and in interaction for as long as science has existed.

The interaction is important. The biological evolutionary process produced information-structured biological systems. The cultural evolutionary process produced methods adapted to reading information-structured systems. But the cultural evolutionary process was itself shaped by the biological evolutionary process, because the scientists who developed and refined those methods were biological organisms whose cognitive capacities had been shaped by biological evolution, and they were reading biological systems whose information structure had been shaped by biological evolution. The methods are adapted to reading biology partly because they were developed by biological minds trying to read biological systems, and the cognitive tools available to those minds, the capacity for algebraic reasoning, the capacity for spatial visualisation, the capacity for experimental logic, were themselves products of biological evolution.

This is the full extension of Dobzhansky's insight. Nothing in biology makes sense except in the light of evolution. And nothing in the history of how we read biology makes sense except in the light of an evolutionary process that operates on methods and minds as well as on genes and organisms. The reading and the thing being read are both products of evolution, and they are matched to each other, the methods to the information, partly because they have coevolved, the cognitive tools of scientists shaping which aspects of biological information become readable, and the structure of biological information shaping which cognitive tools become effective.

AlphaFold makes this coevolution explicit in a way that none of the earlier papers could. AlphaFold reads evolutionary information, the patterns of conservation and covariation in multiple sequence alignments, in order to read structural information, the three-dimensional conformation of a protein. It is a method that uses evolution to read evolution's products. It is, in the most literal sense available to us, evolution reading itself. And the fact that it works, that reading evolutionary patterns in sequence data allows accurate prediction of three-dimensional structure, is itself the deepest demonstration available to us that biological information is structured by evolution into readable forms at every level, from the population to the molecule to the atom.

* * *

The second question you should carry away from this book is whether biology is information at a higher level, which is to say whether the claim that biology is information is more than a metaphor or a useful framing device, whether it is a substantive truth about the nature of biological systems.

After six chapters, the answer is yes, but the yes requires qualification.

The strong version of the claim, that biological systems are nothing but information and that matter and energy are merely the substrate in which the information is implemented, is probably not defensible. Information without a physical substrate is not a coherent concept, and reducing biology to pure information would leave out the physics and chemistry that determine what kinds of information are possible, how information is stored and transmitted, and what errors in transmission look like and how they are corrected. The Ramachandran plot is a demonstration that the physical constraints on protein geometry are not reducible to information. They are constraints set by quantum mechanics and van der Waals physics that determine the space within which biological information can operate. The physics is not secondary to the information. The physics defines the space of possible information.

The defensible version of the claim is more modest but still important and still stronger than most biologists would be comfortable asserting in those terms. It is this: biological systems are systems in which semantic information, information that has consequences for the system that contains it, plays a constitutive role at every level of organisation, from the population to the genome to the protein to the cell. You cannot understand a biological system at any of these levels without understanding the information it contains and processes. And the most powerful methods for studying biological systems are methods for reading that information, for extracting the signal from the noise, at whichever level of organisation is relevant to the question being asked.

Hardy read population-level information. Watson and Crick read molecular structural information. Meselson and Stahl read replication mechanism information. Ramachandran read conformational constraint information. AlphaFold read sequence-to-structure information. Each of these readings revealed something real about the biological world, something that was always there but not visible until the right method was applied at the right level. The information was always in the system. The papers made it readable.

And the reason the information was always in the system, structured and waiting to be read, is evolution. Evolution is the process that creates and maintains biological information at every level of organisation, by selecting for variants whose information is more accurately read, more reliably copied, more effectively used, by the biological machinery that depends on it. Dobzhansky was right. Nothing in biology makes sense except in the light of evolution. What the six papers in this book add to his insight is that the light of evolution illuminates not just the facts of biology but the structure of biological information itself, and not just the structure of biological information but the development of the methods by which we read it.

Biology is information. The information is readable. The readability is an evolutionary product. The methods that read it are adapted to reading it by an analogous evolutionary process. And the most powerful method currently available for reading one of the deepest levels of biological information, AlphaFold, does so by reading the evolutionary history encoded in protein sequence families, which means that it reads biological information by reading evolution, which means that evolution is, through AlphaFold, reading itself.

That is not a metaphor. That is what the six papers, taken together, actually show.

* * *

There is one final thing to say, and it is addressed not to the argument but to you, the person who has read this far.

You began this book with six papers. Some of them were already familiar to you in outline, from your biology classes, from textbook diagrams, from the names attached to equations and principles and plots. What this book has tried to do is show you what was inside those outlines, the confusion and the reasoning and the choices and the social conditions and the disciplinary traditions that produced the clean results you were taught to memorise.

The facts in your textbooks are the residue of science. This book has been about what burned away to leave those facts. The confusion that Hardy dissolved in an afternoon. The model that Watson and Crick built from other people's data. The centrifuge tubes that Meselson and Stahl prepared over months of careful work. The geometry that Ramachandran worked out at the University of Madras with limited resources and enormous precision. The neural network that DeepMind trained on a hundred and seventy thousand protein structures to learn what no human mind could consciously extract from the same data. The historical detective work that Edwards did to understand why something obvious took six years to see.

These are the stories that textbooks do not tell, because textbooks are not written to tell stories. They are written to transmit facts efficiently, which is a legitimate and important purpose but a different purpose from the one this book has served. This book has tried to show you science as a human activity, conducted by specific people in specific places under specific conditions, producing knowledge that is real and reliable but whose production is far messier and more contingent and more interesting than the clean presentation of results in a textbook suggests.

If you have understood what this book was trying to show you, you are now in possession of something that no examination will ask you to demonstrate and no certificate will record. You understand, at least in outline, what scientific thinking actually is, as opposed to what it looks like from the outside when all the confusion has been cleaned away. You understand that the methods scientists use are not arbitrary but are the products of a long process of selection for methods that work. You understand that biological information is not a metaphor but a real property of biological systems at every level of organisation. You understand that the readability of that information is itself an evolutionary product. And you understand that the conversation between the six papers in this book, a conversation conducted across one hundred and thirteen years in the languages of algebra and geometry and experiment and machine learning, is a single argument about the nature of biological knowledge that none of the papers could have made alone.

That understanding is what this book was for. What you do with it is entirely up to you.

* * *

Chapter 7 argues that the readability of biological information is an evolutionary product, that natural selection acting on heritable variation has structured biological systems into information-rich, readable forms over billions of years, and that the scientific methods that read that information are themselves the products of an analogous evolutionary process operating on ideas and methods rather than on genes and organisms.

This argument rests on a claim about natural selection: that it is the mechanism by which biological information is created, structured, and maintained at every level of organisation from the population to the molecule. Chapter 7 states this claim and uses it as the foundation of its argument, but it does not demonstrate it from within biology. It borrows the claim from evolutionary theory and uses it without showing the reader what the claim actually looks like when it is made in biological terms, from biological observation, using biological reasoning.

The final chapter does that. It goes back to the paper, or rather the book, in which the claim was first made in its full generality, in which a human mind working entirely within the conceptual framework of biology, without borrowing any tools from mathematics or physics or computation, arrived at the mechanism that explains why biological information exists and why it has the structure it has. That paper is not a paper but a book, and the book is On the Origin of Species, and the chapter of the book that contains the argument in its most condensed and powerful form is Chapter 4, in which Darwin explains what he means by natural selection and why the three premises of his argument, variation, heredity, and overproduction, make the conclusion, directional change over time, logically unavoidable.

Darwin is the biological foundation of everything this book has argued. He is the last chapter because his argument is the deepest one, the one that explains why all the other arguments were possible, and because a reader who has read the six papers and the synthesis will understand Darwin's argument more completely than a reader encountering it without that preparation. The biology comes last because the biology goes deepest.

Chapter 8

The Naturalist Who Needed No Mathematics

Charles Darwin, On the Origin of Species, 1859

Charles Darwin spent five years on a ship. Between 1831 and 1836, as a young man of twenty-two to twenty-seven, he sailed around the world on HMS Beagle as the ship's naturalist, collecting specimens, making observations, writing notes, and thinking. He was not, at this point, a revolutionary. He was a careful, enthusiastic, somewhat obsessive observer of the natural world, the kind of person who would spend an entire afternoon watching a single species of beetle or a morning examining the barnacles encrusting a rock at low tide, not because he knew what he was looking for but because the habit of close attention to living things was, for him, a form of pleasure that happened also to be a form of inquiry.

By the time the Beagle returned to England in 1836, Darwin had accumulated an enormous quantity of observations, specimens, and notes. He had seen the geological upheaval of South America, the extraordinary diversity of species on the Galapagos Islands, the coral reefs of the Pacific, the fossils of giant mammals in Patagonia that no longer existed anywhere on Earth. He had noticed, on the Galapagos, that the finches on different islands were similar to each other but distinctly different, as if they had all descended from a common ancestor but had diverged in response to the different conditions on each island. He had noticed that the living armadillos of South America resembled, in their general structure, the fossil glyptodonts he had found in the same region, as if the living and the dead were related by descent. He had noticed, throughout the voyage, that species were not distributed randomly across the globe but were clustered in ways that made sense if related species had spread from common points of origin and then diverged as they encountered different conditions.

He did not yet have an explanation for any of this. He had observations, patterns, puzzles. The explanation came later, in England, over twenty years of reading, thinking, experimenting with pigeons and barnacles, and corresponding with naturalists, farmers, and breeders around the world. And when the explanation finally came, it came not from mathematics, not from physics, not from chemistry, but from the accumulated weight of biological observation pressing against a mind that was finally ready to see what the observations were actually saying.

* * *

The argument Darwin eventually published in On the Origin of Species in 1859 is, at its core, a simple argument. So simple that Darwin himself worried it would seem too obvious, that readers would wonder why nobody had seen it before. The argument has three premises and one conclusion, and every premise is a biological observation that any naturalist could confirm.

The first premise is that organisms vary. Within any species, individuals differ from each other in their characteristics. Some are larger, some smaller. Some are faster, some slower. Some are more resistant to particular diseases, some less so. Some have slightly different coloration, or slightly different beak shapes, or slightly different patterns of behaviour. This variation is not random in the sense of being unpatterned. It has a structure, with some variants being more common and others rarer, some variants clustering together and others appearing independently. But the variation is real and pervasive. No two individuals of any species are identical.

The second premise is that variation is heritable. Offspring resemble their parents more than they resemble unrelated individuals. This was not, in 1859, explained at any mechanistic level. Mendel's paper, which would eventually provide that explanation, was published in 1866 and was not read by Darwin. Darwin knew that inheritance happened, that traits ran in families, that breeders could select for desired traits over many generations and reliably produce offspring that expressed those traits more strongly than their parents. He did not know how inheritance worked at the level of physical particles or molecules. But he knew that it happened, because every naturalist and every farmer and every animal breeder knew that offspring resembled their parents, and this was sufficient for his argument.

The third premise is that organisms produce more offspring than can survive. This observation Darwin took from Thomas Malthus, the economist who had noted in 1798 that human populations tend to grow faster than the food supply that supports them, leading inevitably to competition for resources. Darwin extended this observation to all living organisms. Every species produces, over its lifetime, more offspring than will survive to reproduce. Most seeds never germinate. Most insect larvae are eaten before they mature. Most young birds do not survive their first winter. Most offspring, in most species, die before reproducing. This is not an occasional feature of particularly harsh environments. It is a universal feature of all living systems.

From these three premises, the conclusion follows with a logical necessity that Darwin found both compelling and slightly vertiginous. If organisms vary, and variation is heritable, and more offspring are produced than can survive, then the offspring that survive will not be a random sample of the variation present in the population. They will be a non-random sample. The offspring that survive will, on average, be those whose variants are better suited to the conditions they face, better at finding food, better at avoiding predators, better at surviving the diseases and parasites present in their environment, better at competing with other members of their species for the resources available. And because variation is heritable, the offspring of survivors will tend to resemble the survivors, which means the next generation will contain more of the variants that survived and fewer of the variants that did not.

Repeat this process across many generations, and the population changes. It does not change randomly. It changes in a direction, toward variants that are better suited to the conditions the population faces. The change is not designed. No one is choosing which variants survive. The survivors are chosen by the conditions themselves, by the availability of food, the presence of predators, the prevalence of diseases, the competition with other organisms. Darwin called this process natural selection, to distinguish it from the artificial selection practised by breeders who deliberately choose which individuals reproduce. Natural selection is the same logical process as artificial selection, but the selector is not a person with a desired outcome in mind. The selector is the environment, which has no mind and no desired outcome, and which nevertheless filters variation just as effectively as any human breeder.

* * *

This argument is unlike anything in the first six chapters of this book, and it is worth being precise about how it is unlike them.

Hardy's argument was mathematical. It proceeded from algebraic assumptions to algebraic conclusions, and its power came from the precision and generality that mathematics provides. Hardy could prove, in a strict mathematical sense, that his conclusion followed from his premises. If you accept the premises, you must accept the conclusion, because the conclusion is a mathematical consequence of the premises.

Watson and Crick's argument was structural and inferential. It proceeded from experimental constraints, the X-ray data, the base ratios, the known chemistry of the nucleotides, to a proposed structure that satisfied all those constraints simultaneously. The argument was not a proof in the mathematical sense. It was an inference to the best explanation: given all the available constraints, this structure is the most plausible account of what DNA looks like.

Meselson and Stahl's argument was experimental. It proceeded from competing hypotheses through an experimental design to a result that eliminated two of the three hypotheses and confirmed the third. The argument's power came from the clarity of the experimental logic: the design was such that each hypothesis made a different and distinguishable prediction, and the result matched only one of those predictions.

Ramachandran's argument was geometric and deductive. It proceeded from known physical facts about bond geometry and atomic radii through the principle of steric exclusion to the conclusion that most phi-psi combinations were forbidden. Like Hardy's argument, it was a deduction: given the physical facts, the conclusion follows necessarily.

AlphaFold's argument was empirical and computational. It proceeded from a large dataset of known structures through a training process to a predictive model whose accuracy was demonstrated on held-out test cases. The argument was not a proof or an inference to the best explanation in the traditional sense. It was a demonstration: the model works, and its working is evidence that the sequence contains structural information that can be extracted by a sufficiently powerful learning system.

Darwin's argument is none of these things. It is not mathematical, not structural, not experimental in the controlled sense, not geometric, not computational. It is what we might call a naturalistic argument: an argument built entirely from observations of living organisms, reasoned through using only the concepts available to a careful observer of the natural world, arriving at a conclusion that is not a proof, not an inference to the best explanation in the technical sense, not a demonstrated prediction, but something older and in some ways more fundamental than any of these. It is an argument from pattern to mechanism, from the observed regularities of the living world to the process that must be responsible for those regularities, conducted entirely within the conceptual framework of biology itself.

The three premises, variation, heredity, and overproduction, are all biological observations. The logical step from premises to conclusion, the recognition that non-random filtering of heritable variation must produce directional change, is a biological inference. The conclusion, that populations change over time in the direction of better adaptation to their conditions, is a biological claim. No physics is required. No chemistry is required. No mathematics beyond the simplest counting is required. The entire argument is conducted within biology, using biological observations and biological reasoning, to arrive at a biological mechanism.

This is what makes Darwin's argument different from every other paper in this book, and this is what you were pointing toward when you said you wanted a paper that is genuinely and purely biological. Darwin's argument is the proof, if proof is needed, that biology has its own native form of reasoning, a form that does not need to borrow its tools from mathematics or physics or computation but that can, from biological observation alone, arrive at conclusions as foundational and as far-reaching as anything produced by any other mode of scientific thinking.

* * *

Now read the passage from Chapter 4 of the Origin of Species where Darwin summarises the argument for natural selection. It is not a scientific paper in the modern sense. It does not have a Methods section or a Results section or a list of numbered references. It is written in the prose of a Victorian naturalist, careful and precise but also literary, attentive to the reader's possible objections, willing to acknowledge difficulty and uncertainty while pressing forward with the central argument. Read it as you would read the other papers in this book: not for the facts it contains but for the thinking it performs. Notice how Darwin builds the argument from observation to mechanism. Notice which objections he anticipates and how he addresses them. Notice the moments where the argument rests not on a single decisive observation but on the cumulative weight of many observations pressing in the same direction.

And notice, when you have finished reading it, that you have just read the argument that explains why everything else in this book is possible. Hardy could read evolutionary information from allele frequencies because natural selection had structured allele frequencies into patterns that carry information about selection. Watson and Crick could read hereditary information from the double helix because natural selection had preserved, across billions of years, a molecule whose geometry encoded the mechanism of its own reliable copying. Meselson and Stahl could demonstrate the copying mechanism because it was a real mechanism, implemented in real biochemistry, that natural selection had refined to extraordinary fidelity over the entire history of life. Ramachandran could map the allowed conformations of protein backbones because natural selection had filled those conformations with functional proteins, leaving the forbidden regions empty, making the plot informative rather than uninformative. AlphaFold could learn the relationship between sequence and structure because natural selection had produced a world in which related sequences have related structures, a world in which the evolutionary history of a protein family is readable in the covariation patterns of its members.

Darwin's mechanism is the reason the biological world is a world in which information exists, is structured, and is readable. Without natural selection acting on heritable variation in populations that produce more offspring than can survive, there would be no allele frequencies carrying a signal about evolutionary forces, no molecules encoding the mechanism of their own reproduction, no protein sequences specifying their own structures, no evolutionary conservation patterns readable by a machine learning system. There would be no biological information, because there would be no process by which the structure of the environment could be written into the structure of organisms over time.

Darwin is not one paper among six in this collection. He is the paper that explains why the other six are possible. He is the foundation that the conversation has been resting on all along, the argument that was implicit in every chapter but never stated directly until now, the biological reasoning that underlies the mathematical, structural, experimental, geometric, and computational readings that the other papers perform.

* * *

This brings us to the question that the whole book has been building toward, the question you should be in a position to answer now that you have read all seven of the papers and all eight of the chapters.

Is biology information at a higher level?

Darwin's paper provides the final piece of the answer. Not a philosophical piece, not a mathematical piece, not a computational piece. A biological piece.

Information, in the sense that matters for biology, is not just any pattern. It is a pattern that has consequences for the system that contains it. A DNA sequence is information not merely because it is complex or because it can be described mathematically but because it does something, it specifies a protein, it regulates a gene, it determines the development of an organism, and because those consequences affect the survival and reproduction of the organism that contains it. Biological information is semantic information: information that means something to the system that reads it, information whose meaning is constituted by its consequences for survival and reproduction.

Darwin's argument explains why this kind of information exists in biological systems. Natural selection is a process that creates semantic information by linking the structure of organisms to the structure of their environments, through the mechanism of differential survival and reproduction. Variants that are better matched to their environment survive and reproduce more effectively, and their heritable variants are preserved in the population. Over many generations, this process writes information about the environment into the structure of organisms, into their anatomy and physiology and biochemistry and behaviour, in the form of adaptations that make them more effective in that environment.

The allele frequencies that Hardy reads are semantic information: they carry a signal about the selection pressures acting on the population. The DNA sequence that Watson and Crick read is semantic information: it specifies a protein whose structure and function are consequences of billions of years of selection. The protein backbone conformations that Ramachandran maps are semantic information: the allowed regions are populated because organisms whose proteins folded into those conformations survived and reproduced, while organisms whose proteins adopted forbidden conformations did not. The sequence-to-structure relationship that AlphaFold reads is semantic information: the relationship is consistent and learnable because selection has preserved sequences that fold reliably into functional structures, making the mapping between sequence and structure a real biological regularity rather than a coincidence.

Biological systems are systems in which semantic information plays a constitutive role at every level of organisation because natural selection is a process that creates semantic information, that writes the structure of the environment into the structure of organisms, over time and through the mechanism of differential survival and reproduction. Darwin's argument is the biological foundation for the claim that biology is information at a higher level, because it explains how biological information came to exist, why it is structured the way it is, and why it is readable by the methods that the other papers in this book employ.

Biology is information. The information is semantic. The semantics are constituted by the consequences of variation for survival and reproduction. Those consequences are the engine of natural selection. And natural selection, operating on heritable variation in populations that produce more offspring than can survive, is what Darwin described, in biological terms, using biological reasoning, from biological observation, in the book that changed our understanding of life more completely than any other work in the history of science.

* * *

There is one thing left to say, and it is not about Darwin or about biology or about information. It is about the conversation this book has been conducting across one hundred and fifty-eight years, from Darwin's Origin in 1859 to AlphaFold's paper in 2021.

The conversation began with a naturalist on a ship, watching finches on islands and fossils in rock faces and barnacles on ship hulls, accumulating observations without yet having an explanation, waiting for the weight of what he had seen to press him toward a mechanism. It continued with a mathematician who dissolved a biological confusion in an afternoon using arithmetic, a physicist and a zoologist who inferred a molecular structure from X-ray shadows and chemical constraints, two young researchers who watched bands form in a centrifuge tube and saw the copying mechanism of life made visible, a physicist in Chennai who mapped the geometry of what proteins are allowed to be, and a team of engineers who taught a machine to read what billions of years of evolution had written into protein sequences.

Each of these people was working on a specific problem, in a specific place, at a specific time, with specific tools inherited from specific intellectual traditions. None of them knew they were part of a conversation. Hardy did not know that his equation would be used to detect selection in earthworm populations near mining sites in Odisha. Watson and Crick did not know that the copying mechanism they implied in one quiet sentence would be demonstrated by two young researchers in a California laboratory five years later. Ramachandran did not know that his geometric map of protein conformational space would be built into a neural network that would predict protein structures with atomic accuracy sixty years after he drew it. Darwin did not know that the mechanism he described from barnacles and finches and pigeon breeders would turn out to be the process that created all the biological information that the subsequent century and a half of molecular biology would learn to read.

They were not having a conversation. But the papers were. The papers were building on each other, correcting each other, extending each other, reading each other, in the way that scientific papers do across time when they are all trying to understand the same deep thing. And the deep thing they were all trying to understand, from Darwin's mechanism to AlphaFold's predictions, is the same thing: how does a biological system come to contain information about its world, and how can that information be read by a mind that wants to understand it?

Darwin provided the mechanism. The other six papers provided the readings. Together they constitute a single argument, conducted across one hundred and fifty-eight years in the languages of natural history and algebra and geometry and experiment and physics and machine learning, that biology is a world in which information exists, is structured by evolution, and is readable by minds that have themselves been shaped by the same evolutionary process that structured the information they are trying to read.

That is the conversation. It is still going on. You are part of it now.

There is one more thing to say, and it is personal rather than argumentative, which is why it comes here at the end rather than anywhere in the middle.

This book began with a problem I could not solve. A population of plants on an island, flowers of two colours, one dominant over the other, present in certain proportions. What happens after a few generations of random mating? I spent hours on that problem as a BSc student and found nothing. The answer was not available to me because I did not yet have the framework that makes it visible, and nobody had shown me that the framework existed or that finding it was the kind of thing a biology student was supposed to do.

What I did not know at the time was that Dunn and Sinnott had placed that problem there deliberately, before the Hardy-Weinberg chapter, to test whether the reader could arrive at the logic that Hardy and Weinberg had arrived at before being shown the framework. They were practising a pedagogy of withheld answers, trusting that the experience of genuinely not being able to solve a problem would make the solution, when it arrived, land as a discovery rather than a delivery. This book is that same pedagogy practised at a larger scale, across eight texts and one hundred and fifty-eight years, withholding the synthesis until the reading has made it possible to see.

The framework was Hardy-Weinberg equilibrium. When it arrived, the problem dissolved in minutes. The answer had always been there, waiting in the algebra, derivable from three simple assumptions about how alleles combine in random mating populations. I had not been able to see it because I had been given the equation without the question, the result without the reasoning, the surface without the structure beneath.

Darwin's mechanism is the deepest version of that same structure. Natural selection acting on heritable variation in populations that produce more offspring than can survive: three observations, one conclusion, the same logical necessity that Hardy's algebra has, but operating not on allele frequencies in a single generation but on the entire history of life on Earth. The island problem has a clean mathematical answer because Mendelian genetics is true. Mendelian genetics is true because natural selection has preserved the discrete particulate nature of heredity over billions of years, because organisms whose hereditary information was not transmitted reliably did not persist, because the readability of biological information at every level, from the population to the molecule to the protein backbone to the sequence, is the product of the same process that Darwin described from barnacles and finches and pigeon breeders and the fossil record, without any mathematics at all, in the book that changed our understanding of life more completely than any other work in the history of science.

The island problem that defeated me as a BSc student is a Hardy-Weinberg problem. Hardy-Weinberg rests on Mendel. Mendel rests on Darwin. Darwin rests on the patient observation of the living world by a mind willing to follow biological evidence wherever it leads, without borrowing tools from mathematics or physics or computation, trusting the biology to be sufficient.

It was sufficient. It is sufficient. The structure beneath was always there. This book has been an attempt to show you where to look for it, from a degree college in Odisha, by someone who spent hours on an island problem as a student and could not find the answer, and who has been looking for it, in one form or another, ever since.

Afterword

For the Teacher Who Still Reads on Sunday Afternoons

You know who you are.

You are the teacher who finished marking examination papers on Saturday and woke up on Sunday morning and reached for a book that had nothing to do with the syllabus. Not because you were avoiding work. Because the work you were avoiding is not the work that brought you to teaching in the first place. The work that brought you to teaching was something else, something harder to name, a hunger for the structure beneath the facts, a desire to understand not just what is known but how anyone ever came to know it.

You have been teaching for some years now. You know the syllabus. You know which questions appear on the examination and which do not. You know how to explain Hardy-Weinberg equilibrium in a way that your students can reproduce correctly in an answer sheet, and you know that explaining it that way leaves something out, something important, something you have never quite been able to name or find in any book written for teachers like you.

This book was written because of that something.

I am a science teacher at a regional college in Odisha. I have sat where you are sitting. I have felt what you are feeling, the gap between the excitement that brought me to science and the flatness of the way science is usually taught, the distance between the Hardy-Weinberg equation as a formula to be memorised and the Hardy-Weinberg equation as the product of a specific human mind working through a specific confusion on a specific afternoon in Cambridge in 1908. I have wanted, for a long time, to write something that closed that gap, not by making science easier but by making it more visible, by showing the thinking that the results conceal.

That is what this book has tried to do. Whether it succeeded is for you to judge. But I want to tell you, in this final page, what I hope you are feeling now that you have read it.

I hope you are feeling that your intuition was right. That something has been missing from the way science is taught, not just in your classroom but almost everywhere, and that the missing thing is not more content or better pedagogy or newer technology but something older and simpler and harder to provide. The missing thing is the thinking. The confusion before the clarity. The question before the answer. The human being before the result. The structure beneath.

I hope you are feeling that the papers in this book, which you may have known about without having read, or read without having understood in this way, belong to you as much as to anyone. Hardy's algebra was not written for specialists in population genetics. It was written because a friend asked a question that deserved a clear answer. Watson and Crick's quiet sentence was not written for molecular biologists. It was written because two people had found something extraordinary and wanted to point toward it without overclaiming. Ramachandran's geometric reasoning was not produced by someone at a famous institution with abundant resources. It was produced by a physicist in Chennai who found the problem interesting and worked through it with the tools he had. Darwin's argument was not the product of a laboratory or a university or a grant. It was the product of five years of watching the natural world with extraordinary attention and twenty years of thinking about what he had seen.

These papers belong to anyone who reads them carefully. Including you. Including your students.

I hope you are feeling, most of all, that the conversation this book has traced across one hundred and fifty-eight years is not finished. It did not end with AlphaFold in 2021. It is continuing right now, in laboratories and field stations and degree colleges and universities and wherever curious people are sitting with difficult questions and insufficient tools and the stubborn conviction that the structure beneath the surface is there to be found if you look carefully enough.

You are part of that conversation. You have always been part of it. The fact that you are reading this book, on a Sunday afternoon, in a small town somewhere in India or Nigeria or Brazil or Indonesia or anywhere that the major centres of science have not yet paid sufficient attention to, does not place you at the edge of the conversation. It places you exactly where Ramachandran was in Chennai in 1963, and where Hardy was in Cambridge in 1908 when he thought his result was too obvious to be worth publishing, and where Darwin was on the Beagle watching finches on islands and not yet knowing what he was seeing.

The structure beneath is visible from wherever you are standing. You just have to know where to look.

This book has tried to show you where to look. What you do with that is entirely up to you. But I suspect, if you have read this far, that you already know what you are going to do.

You are going to go back to your classroom on Monday morning and teach differently.

Not because this book told you to. Because your intuition has been telling you to for years, and now you have found, in these eight texts and the thinking behind them, the words for what your intuition already knew.

That is enough. That is everything.

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The Eight Texts

These are the primary texts this book reads. All of them can be found online, most of them freely. The note after each entry tells you where to look.

One. G. H. Hardy, "Mendelian Proportions in a Mixed Population," Science, New Series, Volume 28, July 10, 1908, pages 49 to 50. Freely available through JSTOR and through the Yale Journal of Biology and Medicine reprint of 2003, which is the version most easily found online. Search for Hardy 1908 Mendelian Proportions and it will appear within the first few results. It is two pages long. Read it before reading Chapter 1, and then read it again after.

Two. A. W. F. Edwards, "G. H. Hardy (1908) and Hardy-Weinberg Equilibrium," Genetics, Volume 179, July 2008, pages 1143 to 1150. Freely available through the Genetics Society of America's open access archive at genetics.org. This is the paper that tells the story behind Hardy's paper, and it should be read alongside Hardy's original rather than instead of it.

Three. J. D. Watson and F. H. C. Crick, "Molecular Structure of Nucleic Acids: A Structure for Deoxyribose Nucleic Acid," Nature, Volume 171, April 25, 1953, pages 737 to 738. Freely available through Nature's archive and through multiple educational websites that have reproduced it with permission. It is 900 words. Read it slowly. Read the last paragraph twice.

Four. M. Meselson and F. W. Stahl, "The Replication of DNA in Escherichia coli," Proceedings of the National Academy of Sciences, Volume 44, Number 7, July 1958, pages 671 to 682. Freely available through PubMed Central at ncbi.nlm.nih.gov/pmc. Search for Meselson Stahl 1958 Replication DNA. The figures in this paper are worth studying carefully before reading the text, because the figures are where the experimental logic is most clearly visible.

Five. G. N. Ramachandran, C. Ramakrishnan, and V. Sasisekharan, "Stereochemistry of Polypeptide Chain Configurations," Journal of Molecular Biology, Volume 7, 1963, pages 95 to 99. This paper requires institutional access through Elsevier, but it can be found through university library portals and through academic sharing platforms. It is more technically demanding than the other papers in this collection, and the reader who finds the notation unfamiliar should read Chapter 5 first and then return to the paper with the chapter's framework in mind.

Six. John Jumper, Richard Evans, Alexander Pritzel, and colleagues, "Highly Accurate Protein Structure Prediction with AlphaFold," Nature, Volume 596, August 26, 2021, pages 583 to 589. Freely available as open access through Nature at nature.com. The supplementary methods, which run to several hundred pages, are not required reading for the argument of Chapter 6, though they are remarkable as a document of the scale and complexity of the work. The main paper is sufficient and rewarding on its own.

Seven. A. W. F. Edwards, already listed above as Two.

Eight. Charles Darwin, On the Origin of Species by Means of Natural Selection, or the Preservation of Favoured Races in the Struggle for Life, John Murray, London, 1859. First edition. Freely available in its entirety through Project Gutenberg at gutenberg.org, through the Darwin Online archive at darwin-online.org.uk, which includes facsimile scans of the original edition, and through many other digital archives. Chapter 4 of the Origin, titled Natural Selection, contains the core argument discussed in Chapter 8 of this book. Read Chapter 4 of the Origin alongside Chapter 8 of this book, and then, if you find yourself wanting more, read the rest of the Origin at whatever pace suits you. It rewards a slow and patient reading.

A note on access: several of these papers require institutional access that not every reader will have. University and college libraries in India increasingly provide access to major journals through the INFLIBNET N-LIST programme, which covers most of the journals listed above. Readers without institutional access should contact their nearest college or university library, which can often obtain papers through interlibrary loan. The argument that scientific knowledge should be freely available to anyone who wants to read it is one this book endorses entirely.

Going Further

These are books and papers that extend the arguments of specific chapters. The list is short and deliberate. Every item on it is worth your time.

On Hardy and the history of population genetics

William B. Provine, The Origins of Theoretical Population Genetics, University of Chicago Press, 1971. This is the most complete account of how Hardy-Weinberg equilibrium was eventually integrated into the modern synthesis that united Darwinian natural selection with Mendelian genetics. Provine traces the story from Mendel's rediscovery in 1900 through the work of Fisher, Wright, and Haldane in the 1920s and 1930s, and shows how the mathematical foundations of population genetics were built on the framework Hardy established in two pages in 1908. It is a model of intellectual history.

James F. Crow, "Eighty Years Ago: The Beginnings of Population Genetics," Genetics, Volume 119, 1988, pages 473 to 476. A short, beautifully written essay by one of the most distinguished population geneticists of the twentieth century, reflecting on what Hardy and Weinberg actually established and what it took for the field to absorb it. Crow writes with the clarity of someone who has thought about these questions for a lifetime.

On Watson, Crick, and Franklin

Horace Freeland Judson, The Eighth Day of Creation: Makers of the Revolution in Biology, Cold Spring Harbor Laboratory Press, expanded edition 1996. This is the most complete account of the intellectual world of molecular biology from the 1940s through the 1970s, built from interviews with almost everyone involved. Judson interviewed Watson, Crick, Franklin's colleagues, Meselson, Stahl, and dozens of others over many years, and the result is the closest thing we have to a full picture of what it was actually like to be doing molecular biology in that period. It is long and it is essential.

Brenda Maddox, Rosalind Franklin: The Dark Lady of DNA, HarperCollins, 2002. The most thorough biography of Franklin available, written with access to her letters and notebooks. Maddox does not attempt to make Franklin into a simple victim or a simple hero. She shows a complex, brilliant, difficult, and generous person whose contribution to the discovery of the double helix was real and whose recognition during her lifetime was inadequate. Read it alongside Chapter 3.

On Meselson, Stahl, and experimental design

Frederic Lawrence Holmes, Meselson, Stahl, and the Replication of DNA: A History of the Most Beautiful Experiment in Biology, Yale University Press, 2001. Holmes reconstructed the experiment in extraordinary detail from laboratory notebooks, correspondence, and interviews with Meselson and Stahl themselves. The result is a book-length account of how a single experiment was conceived, designed, performed, interpreted, and eventually recognised as definitive. It is the most detailed case study of experimental scientific thinking available for any experiment in biology.

On Ramachandran

G. N. Ramachandran and V. Sasisekharan, "Conformation of Polypeptides and Proteins," Advances in Protein Chemistry, Volume 23, 1968, pages 283 to 437. This is the longer and more comprehensive treatment of the work introduced in the 1963 paper that Chapter 5 discusses. It is technically demanding but it gives a fuller picture of the geometric reasoning behind the Ramachandran plot and of the broader programme of structural research that Ramachandran was pursuing at the University of Madras.

Jane S. Richardson, "The Anatomy and Taxonomy of Protein Structure," Advances in Protein Chemistry, Volume 34, 1981, pages 167 to 339. Richardson's paper is the work that made protein secondary structure visually comprehensible to a generation of structural biologists, through the ribbon diagrams she invented that are now the standard way of depicting protein structure. Reading it alongside the Ramachandran paper shows how Ramachandran's geometric framework shaped the visual language of protein biochemistry for the subsequent half century.

On AlphaFold and the nature of computational knowledge

Kathryn Tunyasuvunakool and colleagues, "Highly Accurate Protein Structure Prediction for the Human Proteome," Nature, Volume 596, August 26, 2021, pages 590 to 596. This is the companion paper to the AlphaFold paper, published in the same issue of Nature, reporting the application of AlphaFold to the entire human proteome. Reading it alongside the main AlphaFold paper gives a clearer picture of what it means in practice to be able to predict protein structures at scale.

Mohammed AlQuraishi, "AlphaFold at CASP13," Bioinformatics, Volume 35, Number 22, 2019, pages 4862 to 4865. AlQuraishi is one of the most thoughtful commentators on AlphaFold and its implications. This paper, written after AlphaFold's first CASP entry in 2018, raises many of the epistemological questions that Chapter 6 of this book explores, and it raises them from the perspective of a working computational biologist rather than a philosopher of science.

On Darwin

Janet Browne, Charles Darwin: Voyaging, Princeton University Press, 1995, and Charles Darwin: The Power of Place, Princeton University Press, 2002. This two-volume biography is the most complete account of Darwin's life and the development of his thinking available in English. The first volume covers his life through the voyage of the Beagle and the years of private theorising that followed. The second covers the publication of the Origin and the remaining years of his life. Browne is a historian of science who writes with remarkable clarity and depth. If you want to understand how the argument of the Origin came to be made, these two volumes are the place to go.

Peter Godfrey-Smith, Philosophy of Biology, Princeton University Press, 2014. This is the most accessible treatment of the philosophical questions that Darwin's argument raises, written by a philosopher who is also a trained marine biologist. Godfrey-Smith addresses the question of what natural selection actually is, whether it is a law or a principle or something else entirely, what it means for evolution to produce information, and how biological explanation differs from physical explanation. It is an ideal companion to Chapter 8 of this book and to Chapter 7's argument about the relationship between evolution and the readability of biology.

On scientific thinking as a subject

Thomas S. Kuhn, The Structure of Scientific Revolutions, University of Chicago Press, 1962. Third edition 1996. Kuhn's argument that science develops not by steady accumulation but through periodic revolutionary restructurings of its fundamental frameworks is the most influential work in the philosophy of science of the twentieth century, and it is directly relevant to several of the stories this book tells. Hardy's dissolution of the dominant-trait misconception, Watson and Crick's replacement of protein as the genetic material with DNA, AlphaFold's replacement of first-principles simulation with learned pattern recognition, all of these can be read as instances of the kind of paradigm shift Kuhn described. Whether Kuhn's account is fully correct is debated, but reading it will give you a framework for thinking about the episodes this book narrates.

David L. Hull, Science as a Process: An Evolutionary Account of the Social and Conceptual Development of Science, University of Chicago Press, 1988. Hull's argument that science itself can be understood through an evolutionary framework, with ideas and methods varying, being selected, and being transmitted through scientific communities in ways that parallel biological evolution, is the most sustained treatment of the idea that Chapter 7 of this book develops. Hull is more technical and more contentious than Kuhn, but for the reader who wants to pursue the evolutionary analogy seriously, this is the book to read.

Acknowledgements of Debt

Every book is built on other books, and this one is no exception. The debts this book owes are intellectual rather than personal, owed to thinkers whose work shaped the argument without always being cited in the text. This page acknowledges the most important of those debts.

The central argument of Chapter 7, that scientific methods are themselves products of a cultural evolutionary process that operates analogously to biological natural selection, owes its clearest formulation to David Hull's Science as a Process, which the previous list recommends. Hull spent a career developing the idea that scientific communities are the units of selection in the evolution of scientific knowledge, and that concepts and methods compete, vary, and are differentially transmitted in ways that parallel the competition, variation, and differential reproduction of organisms. This book's version of the argument is less technical than Hull's and more directly connected to the specific papers it discusses, but the intellectual debt is real.

The extension of Dobzhansky's insight that Chapter 7 proposes, the claim that not only does nothing in biology make sense except in the light of evolution, but nothing in the history of how we read biology makes sense except in the light of an analogous evolutionary process operating on scientific methods, is the book's own contribution. But it could not have been made without Dobzhansky's original 1973 essay, which remains worth reading in full. Theodosius Dobzhansky, "Nothing in Biology Makes Sense Except in the Light of Evolution," The American Biology Teacher, Volume 35, Number 3, March 1973, pages 125 to 129.

The argument that biological information is semantic rather than merely syntactic, that it is information that has consequences for the systems that contain and read it rather than just complexity that can be described mathematically, draws on a tradition in the philosophy of biology that includes Peter Godfrey-Smith's work on Darwinian populations and Kim Sterelny and Paul Griffiths's Sex and Death: An Introduction to Philosophy of Biology, University of Chicago Press, 1999. The specific formulation used in this book is the author's own, but the distinction between syntactic and semantic information as it applies to biology is not original here.

The decision to include G. N. Ramachandran as a central figure in a book about landmark texts in biology, rather than treating him as a technical footnote to the

AlphaFold story, reflects a conviction that the history of biology as it is usually told systematically underrepresents contributions made from outside the major Western scientific centres. This conviction is shared by a growing number of historians of science, and the author's thinking has been shaped by reading in the history of science in postcolonial contexts, including Projit Mukharji's Nationalizing the Body and Dhruv Raina and S. Irfan Habib's work on the history of science in India. These works are not directly cited in the main text but they are part of the intellectual formation that made the Ramachandran chapter possible.

Finally, this book owes a debt to the teachers who first showed the author that a scientific paper is not a source of facts to be extracted but a record of a human mind at work. That debt cannot be repaid by citation. It can only be repaid by writing a book that tries to do for others what those teachers did, which is to say: look here, look carefully, the thinking is visible if you know where to look.

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